

Evaluating the effectiveness of open-source LLMs for multi-domain Bengali text classification with zero-shot and few-shot prompting

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Abstract

Large Language Models (LLMs) have advanced multilingual NLP significantly, yet their effectiveness for Bengali, a globally important but low-resource language, remains understudied due to limited linguistic resources and the absence of a standardized evaluation benchmark. This is due to the unavailability of large linguistic resources and the lack of a standard benchmarking framework, which has yet to be established. This study aims to address the underrepresentation of Bengali text classification by leading a systematic evaluation. It involves assessing multiple open-access LLMs on Bengali text classification tasks and observing their limitations and regular error patterns. The procedure consists of six open-source LLMs, which include LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B, and are evaluated using four critical text classification tasks in Bengali, namely sentiment analysis, emotion recognition, hate-speech detection, and fake-news classification. Furthermore, two prompting methods are compared in the context of zero-shot and few-shot prompting using the vLLM model to investigate the adaptability of the model in resource-limited settings. The outcomes of our study indicate that, regardless of the dataset, the bigger multilingual models always performed better in comparison with the smaller ones. When training in zero-shot, Qwen-2.5-72B showed the best performance at 79.90% accuracy and 79.88% **Macro-F1** score, showing amazing accuracy on hate-speech detection. In the case of fake news, Gemma-2-27B performed best at 84.45% accuracy and 84.33% **Macro-F1** score. With a few-shot set-up, Gemma-2-27B and Qwen-2.5-72B demonstrated a high level of performance in terms of **Macro-F1** score, reaching 84.17% for fake-news detection and 78.25% for emotion recognition, respectively. After the final observation, the study reveals that the model size, quality of multilingual pretraining, and contextual samples have a vital impact in low-resource settings.

Introduction

Large Language Models (LLMs) have revolutionized natural language processing.
Large Language Models (LLMs) facilitate enhanced efficiency on different types of

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tasks, without large-scale training related to the task. The models utilize context-based learning to comprehend the pattern from the given prompts, which in turn decreases the requirement for a large labelled dataset. Nevertheless, it is notable that most achievements of LLMs are observed in high-resource languages as English, while the low-resource languages like Bengali do not receive enough attention. Bengali is a rich language with immense diversity, even though it is considered to be the 7th most spoken language in the world, and there are over 230 million Bengali speakers; this widely spoken language is still underrepresented [1]. In [2, 3], authors mention that exclusive-use licensed models like GPT-4 and Claude-3.5 present moderate capabilities in processing the Bengali language. On the other hand, publicly accessible models like Gemma-2 and LLaMA-3.1 show disappointing outcomes in the same task.

The main drawbacks are data quality and the substandard performance of well-known robust LLMs. The majority of the Bengali language processing conducted previously depends heavily on the machine-generated translated datasets like Bangla-Alpaca or Bangla-Orca translated version of the original Stanford Alpaca and OpenOrca dataset [4], hence it causes a rise in various errors and poor performance from the models [5]. Moreover, an extensive review shows that well-known robust LLMs encounter difficulties just to reach satisfactory outcomes in Bengali tasks. BenLLM-Eval finds that from time to time, GPT-3.5 and Claude models' performances match the performance of fine-tuned Bengali models on a few tasks [6]. Furthermore, in the case of zero-shot, the performance of the majority of LLMs is substantially inadequate, and publicly accessible models (like LLaMA-2-13B) perform worse.

In case of accuracy, sizeable differences are observed between English and Bengali text-related tasks, which is more significant for smaller model series (e.g., Mistral) [7]. This difference in accuracy can be overcome via in-context learning. [8] shares that by presenting a few suitable samples, the LLMs showcase significantly enhanced understanding of the low-resource languages. Nonetheless, the research comprising Bengali language-related LLMs is still scarce.

Recently, researchers have taken the initiative to resolve the accuracy difference that LLMs face by building exclusive Bengali language resources. [9] presented the first substantial pretrained Bengali LLMs named TituLLMs. The authors also introduced five new Bengali baseline datasets and showed that the performance of Bengali-specific models is far better than that of other language models for native tasks. TigerLLM introduced in the study [2], trained on a meticulously refined Bengali dataset, showcases a new benchmark for Bengali language processing, surpassing the previous publicly accessible models. GPT-3.5 Turbo achieved more than 97% accuracy in Bengali hate speech detection by utilizing few-shot prompting, which outperforms streamlined transformer models [10]. Regardless of the mentioned accomplishments, there is still a limited number of extensive multi-domain baseline and systematic prompt assessments. The majority of previous research focuses on individual tasks or utilizes translated English baselines, which led to a rise in noise [5, 7]. Confronting these differences in outcomes is vital for creating a multilingual natural language processing model. The main objective of our research is to provide a systematic evaluation of publicly accessible LLMs based on four Bengali text classification tasks under zero-shot and few-shot prompting conditions.

According to a few recent studies, model dimension and the scope of the training dataset have an immense influence on the downgrade of performance [11]. In a study [12], bigger models show amazing performance in zero-shot tasks for high-resource languages like English; however, the performance plummets for low-resource languages. And many LLMs can learn and solve problems with the help of few-shot prompting [13]. [8] reveals that introducing a few samples via few-shot prompting helps enhance the model performance across 25 undervalued languages.

However, the Bengali language is not specifically included in this list of undervalued languages. Because of the lack of large-scale pre-trained LLMs focused on the Bengali natural language processing domain, most advancements rely heavily on streamlined models like BanglaBERT [14]. During the research, [7] found that current multilingual datasets consist of around 30 billion Bengali text tokens; in contrast, the English language consists of a trillion tokens, which results in poor model quality. In [6], build BenLLM-Eval, a baseline model capable of summarization, classification, and QA tasks in the Bengali language. Additionally, they notice that even though GPT-3.5 sometimes outperforms the fine-tuned models, its performance is poor for a zero-shot system. Likewise, [9] introduced TituLLMs and native baselines, showcasing enhanced performances across multilingual models. Regardless of these initiatives, significant research gaps are still present. Bengali language classification has not been analyzed appropriately via zero-shot and few-shot settings. Latest publicly accessible models, namely Qwen, Mistral, or Phi-4, have yet to be evaluated on Bengali text classification tasks, and overall error assessment of the models is lacking. To tackle these research gaps, our study aims to provide a systematic evaluation of six publicly accessible LLMs across four Bengali text classification tasks with further insight into their limitations and error patterns.

To analyze the performance of publicly accessible LLMs for Bengali text classification tasks, this study is guided by the following research questions:

- **RQ1:** What is the performance of publicly accessible LLMs on Bengali text classification over several different domains, consisting of sentiment analysis, emotion recognition, hate speech detection, and fake news detection?
- **RQ2:** What is the impact of model size and pretraining of multilingual models on text classification performance, specifically for low-resource languages like Bengali?
- **RQ3:** Is there any noteworthy improvement noticed in accuracy, recall, and F1-score when utilizing few-shot prompting, in contrast to zero-shot prompting, in the Bengali text classification task?
- **RQ4:** What sort of incorrect classification, limitations, or systemic errors are usually found during LLMs performance for Bengali text tasks, and what type of limitations are stipulated by these error patterns?
- **RQ5:** How uniform are various publicly accessible LLMs in classifying diverse Bengali natural language processing tasks, when assessed using a unified, prompt-based pipeline?

This research presents several noteworthy contributions to the rising field of the Bengali natural language processing domain:

1. **Comprehensive Evaluation:** We perform a comprehensive evaluation of six of the latest publicly accessible LLMs: LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B, on four types of Bengali text classification tasks: sentiment analysis, emotion recognition, hate speech detection, and fake news classification.
2. **Unified Prompting Framework:** We apply a systematic evaluation pipeline utilizing the vLLM structure, while using zero-shot and few-shot prompting to maintain consistency and reproducibility over the models' experiments.
3. **Performance Insights:** The overall outcome highlights the significantly better performance of the larger multilingual models (like Qwen-2.5-72B, Gemma-2-27B), specifically in complicated tasks (like emotion recognition), in contrast to smaller models. This shows the significance of model scale and multilingual model training.

4. **In-Context Learning Benefits:** We realized that few-shot prompting increases average recall by over 6%, exhibiting that context-based learning samples improve the model’s understanding of low-resource languages like Bengali.
5. **Detailed Error and Class-Wise Analysis:** We utilized comprehensive evaluation metrics (Accuracy, Precision, Recall, F1-score) and confusion matrices. This enables recognizing common model errors and recommending an approach for future improvements.

The paper is organized into the following sections: First, the Related Work section presents a literature review of prior research on Large Language Models (LLMs), multilingual natural language processing, and Bengali text classification. The Methodology section provides a detailed explanation of the chosen models, dataset preparation, experiment structure, and application of zero-shot and few-shot prompting. The Results section reports the overall performances of the models using evaluation metrics for four types of Bengali text classification tasks, and also presents a comparison among the chosen models. The Discussion section provides detailed insights into the performance patterns, overall findings, common mistakes, and errors related to misclassification. Lastly, the Conclusion section summarizes the contribution of the research, notes the limitations, and recommends future direction of work.

Related Work

Large Language Models (LLMs) for the Bengali language are currently gaining growth, similar to previous advances in Natural Language Processing. Though Bangla is one of the most widely spoken languages in the world, it still lags behind in the field of computational linguistics due to limited annotated corpora, ineffective tokenization, and inadequate domain coverage. In recent times, the focus has been on model adaptation, resource generation, and zero- or few-shot prompting strategies to improve Bengali text understanding and classification across multiple domains.

Progress in Bengali LLMs and Benchmark Development

Recent years have witnessed substantial progress in Bengali natural language processing (NLP), driven by the development of Bengali-specific pre-trained language models, multilingual large language models (LLMs), benchmark datasets, and evaluation frameworks. Table 1 summarizes representative studies and their key contributions to Bengali NLP research.

As summarized in Table 1, prior studies have explored Bengali-specific pre-trained language models, multilingual LLM evaluation, benchmark construction, prompting strategies, and task-specific applications such as sentiment analysis, hate speech detection, and question answering. These efforts collectively demonstrate the growing maturity of Bengali NLP research while also revealing persistent challenges related to tokenization, contextual understanding, and culturally grounded language phenomena. Motivated by these developments, the present study provides a comprehensive evaluation of six contemporary open-source LLMs across four Bengali text classification tasks under both zero-shot and few-shot prompting settings.

Resource Creation and Cross-Domain Studies

Mahtab *et al.* [22] developed BanNERD, a comprehensive 29-domain Bengali NER dataset intended to improve contextual generalization in named-entity recognition

Table 1. Summary of recent studies on Bengali LLMs, benchmarks, and text classification.

Reference	Model	Task / Domain	Key Findings
Dehan et al. [15]	Gemma-2B, TinyLLaMA, Falcon, BanglaBERT	Bengali text classification	TinyLLaMA achieved the best performance for fake news detection, while BanglaBERT excelled in emotion recognition.
Rahman et al. [16]	BERT, RoBERTa	Sentiment analysis	Achieved 88% accuracy on Bengali movie reviews, demonstrating the superiority of transformer-based models over traditional machine learning approaches.
Kabir et al. [6]	BenLLM-Eval	Bengali LLM evaluation benchmark	Introduced a benchmark framework for evaluating GPT-3.5, Claude-2, and LLaMA-2 across diverse Bengali NLP tasks.
Bhattacharjee et al. [14]	BanglaBERT	Bengali language modeling	Developed a Bengali-specific language model trained on 27 GB of text and demonstrated strong performance in sentiment analysis and NER.
Mahmud et al. [17]	Hybrid BanglaBERT framework	Sentiment analysis	Combined lexicon-based scoring with BanglaBERT embeddings and achieved an F1-score of 93%.
Bhowmik et al. [7]	Qwen, Mistral, DeepSeek	Bengali benchmark evaluation	Reported that tokenization issues and translation noise remain major obstacles for multilingual LLMs.
Al Nazi et al. [18]	GPT-4o, Gemini, BLOOM	Zero-shot and few-shot evaluation	Compared open- and closed-source models and highlighted the rapid progress of prompting-based approaches.
Prome et al. [19]	Metaphor-based prompting	Hate speech detection	Improved interpretability and fairness in Bengali hate speech classification through metaphor-guided prompting.
Banda et al. [20]	Few-shot multilingual LLMs	Agricultural question answering	Demonstrated strong generalization capabilities under limited supervision.
Raihan and Zampieri et al. [2]	TigerLLM	Bengali instruction-tuned LLM	Released an open-source Bengali LLM family with performance comparable to GPT-3.5 on several tasks.
Mahfuz et al. [21]	Bengali-specific LLM analysis	Bengali NLP	Highlighted the importance of language-specific pre-training for capturing Bengali linguistic and cultural nuances.

algorithms. In a similar vein, Shafayat *et al.* [23] published BEnQA, a bilingual Bengali-English question-answering dataset that demonstrated considerable thinking differences between the two languages. Singh *et al.* [24] further multilingual research by developing IndicGenBench, a generative benchmark encompassing 11 Indic languages, indicating that contemporary multilingual systems still face significant hurdles in addressing morphological and syntactic variances. Shukla *et al.* [25] introduced IndicMMLU-Pro, a multilingual reasoning benchmark, and found that Indic LLMs continue to lag English-centric models in comprehension tests. Expanding resources for Bengali NLP. Fahim *et al.* [26] presented BanglaTLit, a transliteration dataset with 42.7K Romanized Bengali items, in which fine-tuned encoder models reached cutting-edge performance. Haider *et al.* [27] introduced BANTH, a multi-label hate-speech dataset for transliterated Bengali that significantly increased zero-shot classification accuracy. Romim *et al.* [28] assembled BD-SHS, a dataset of 50,000 online hate-speech samples, and achieved a 91% F1-score using contextual learning approaches. Nayeem *et al.* [29] proposed MultiBanFakeDetect, a multimodal approach for detecting false news that combines textual and visual data to improve cross-domain robustness. Furthermore, Hasan *et al.* [30] developed Bangla_ABSA, the first aspect-based sentiment analysis dataset for Bengali, which covers domains such as restaurants, mobile devices, and autos, allowing for fine-grained polarity recognition.

Nahin *et al.* [9] continued the advancement in large-scale Bengali modeling by

introducing TituLLMs, a family of Bengali large language models trained on 37 billion tokens that outperformed multilingual models such as LLaMA and Zephyr in reasoning and comprehension tasks. Faria *et al.* [10] compared GPT-3.5 Turbo and Gemini 1.5 Pro to transformer-based baselines for hate-speech recognition and achieved 98.5% accuracy in a few shots. Hasan *et al.* [31] explored zero-shot and few-shot prompting for sentiment analysis and discovered that in-context examples greatly improve LLMs’ generative capacities. Rafid *et al.* [32] compared decoder-only and encoder-based models for multi-label document categorization and concluded that decoder architectures respond better to unstructured textual input. Likewise, Saha *et al.* [33] established a hybrid knowledge-driven model that identifies altered Bengali news using language signals and semantic embeddings, with an accuracy of more than 90%.

Domain-Specific LLM Adaptations and Applications

Hossain *et al.* [34] proposed BanglaNewsClassifier, a hybrid stacking ensemble that uses SVM, Random Forest, and XGBoost models to classify 118K Bengali news articles with 94% accuracy. Chowdhury *et al.* [35] demonstrated DepGPT, a fine-tuned large language model designed to detect depressive language in social media content, with an accuracy of 97.9%, demonstrating how LLMs may efficiently capture emotional and psychological patterns in text. Kamruzzaman *et al.* [5] generated BanMANI, an annotated dataset for detecting altered Bengali social media news, underlining the significance of domain-specific data resources for misinformation identification. Faria *et al.* [3] evaluated the performance of large language models (GPT-3.5, Gemini 1.5 Pro) and transformer-based architectures (BanglaBERT, DistilBERT) on Bengali natural language inference, revealing significant variations in reasoning capacities. Their findings showed that few-shot prompting considerably improved reasoning abilities over standard fine-tuning methods. Maity and Deroy *et al.* [36] investigated prompt-tuning strategies for Bengali grammatical error explanation using GPT-4, GPT-3.5, and LLaMA-2 and found significant increases in both accuracy and interpretability, indicating a potential direction for educational NLP applications in Bengali.

Overall, this research show Bengali NLP’s rapid progress in model construction, dataset development, and cross-domain evaluation. Earlier works were primarily concerned with resource construction and baseline transformer models; however more current research has switched to rapid engineering, multilingual adaptability, and task-specific implementations. Despite these developments, Bengali NLP continues to confront issues like as data asymmetry, inadequate tokenization, and a lack of fine-tuning resources. Building on these advances, the current study conducts a thorough evaluation of open-source LLMs for multi-domain Bengali text categorization under zero-shot and few-shot prompting scenarios, revealing novel insights into scalable and resource-efficient modelling for low-resource languages.

Methodology

This research performs four types of Bengali text classification utilizing six open-source Large Language Models (LLMs). The six LLMs include LLaMA-3.2-3B (Meta AI), Mistral-V3-7B (Mistral AI), DeepSeek-R1-8B (DeepSeek), Phi-4-14B (Microsoft), Gemma-2-27B (Google DeepMind), and Qwen-2.5-72B (Alibaba Cloud). The four Bengali text classification tasks were sentiment analysis, emotion recognition, hate speech detection, and fake news detection.

The selected LLMs represent a diverse range of model sizes, pre-training corpora, and post-training optimization strategies. Although all evaluated models are based on

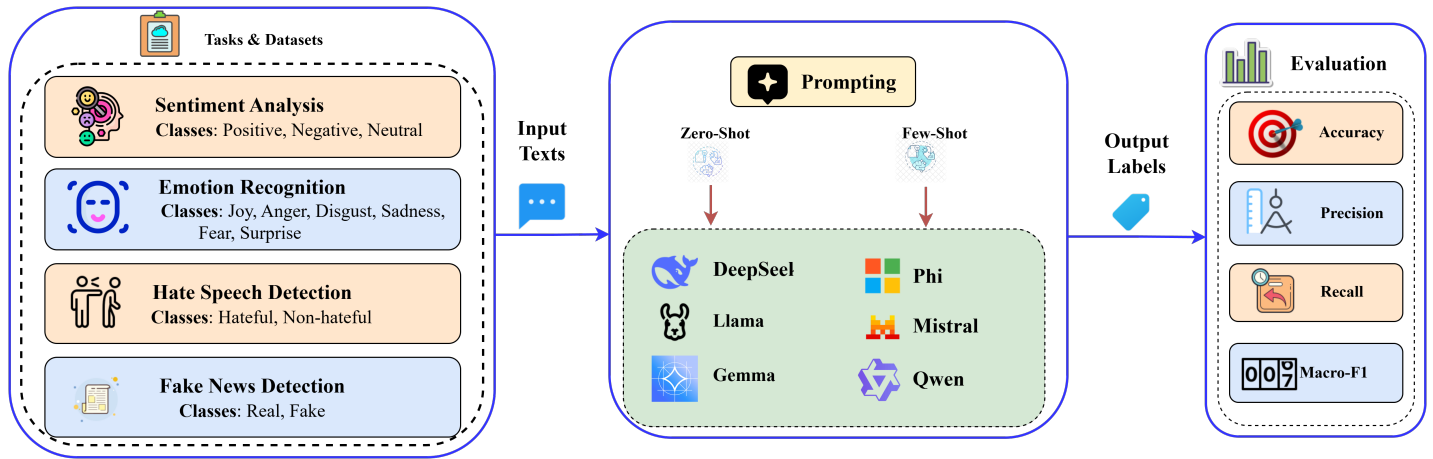


Fig 1. Overall methodology illustrating the evaluation framework for six open-source LLMs across four Bengali text classification tasks under zero-shot and few-shot prompting

Table 2. Summary of datasets used in this study.

Task	Dataset	Classes	Test Size (Samples)
Sentiment Analysis	SentiNoB [37]	3	1,586
Emotion Recognition	BEmoC [38]	6	625
Hate Speech Detection	Bengali HS [39]	2	1,000
Fake News Detection	BanFakeNews [40]	2	508

Transformer decoder architectures, they differ in parameter scale, multilingual capabilities, instruction-tuning procedures, and alignment techniques. Some models additionally employ reinforcement learning-based post-training to enhance reasoning and instruction-following performance. This diversity enables a comprehensive comparison of open-source LLMs for Bengali language understanding and text classification.

The methodology of this research is presented in the diagram, Fig.1. Initially, the Bengali text classification datasets were prepared from multiple domains, which include sentiment analysis, emotion recognition, hate speech detection, and fake news detection. The models separately undergo a zero-shot and a few-shot experimental setup. It mainly helps understand the models' in-context learning ability. The models were provided with the input sample and task instructions in case of the *zero-shot* setup. And for the *few-shot* setup, before the test input, five labeled Bengali text samples were added to the prompt, enabling the model to determine contextual relationships. The purpose of this setup is to assess whether the sample-based context enhances the performance for low-resource languages.

Datasets and Tasks

This research aims to evaluate Large Language Models based on their performance in Bengali text classification. To properly assess the models, four types of text classification were divided into two categories: opinion-based judgment (sentiment analysis and emotion recognition) and fact-based verification judgment (hate speech detection and fake news detection). The detailed information on the datasets is displayed in Table 2.

Sentiment Analysis

The SentiNoB [37] dataset was used for sentiment analysis. This dataset consists of three categories collected from social media and annotated into positive, negative, and neutral. The categories contain 1586 samples, where positive was 654, negative was 571, and neutral was 361.

Emotion Recognition

The BEmoC dataset [38] was used for emotion recognition analysis. The dataset was annotated into six categories: joy, anger, disgust, sadness, surprise, and fear. As usual, this dataset was gathered from social networking sites. The test set comprises 625 samples.

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Hate Speech Detection

The Bengali Hate Speech Dataset [39] was used to evaluate the hate speech detection capability of models. The binary classification framework was implemented for uniformity. A balanced dataset proportion can be viewed in the test set, which consists of 500 samples of hateful and 500 samples of non-hateful samples.

Fake News Detection

Fake news detection was assessed using the BanFakeNews dataset [40]. The dataset consists of approximately 20,000 real news articles and 254 fake news articles. Since the dataset is highly imbalanced, the majority class (real news) was randomly downsampled using a fixed random seed of 42 to ensure reproducibility. Specifically, all 254 fake news samples were retained, and 254 real news samples were randomly selected from the majority class, resulting in a balanced evaluation set of 508 samples with a 1:1 class ratio. We avoided oversampling methods such as text duplication or synthetic data generation to prevent introducing artificial patterns into the evaluation. Since the evaluation set was constructed solely from existing samples, no textual content was modified during the data preparation process. However, we acknowledge that random downsampling may not fully preserve the semantic distribution of the original dataset, which represents a potential limitation of this evaluation setting. Despite this limitation, the balanced test set provided a practical and consistent setting for comparing the performance of different LLMs under identical evaluation conditions.

Models

In this research, experiments were conducted on six Large Language Models (LLMs). The models were selected based on variations in model size, training data, and optimization strategies. The evaluated models range from smaller 3B-parameter systems to larger multilingual 72B-parameter systems. Although all evaluated models

Table 3. Overview of evaluated large language models.

Model	Parameters	Architecture	Developer
LLaMA-3.2-3B	3B	Transformer	Meta AI
Mistral-V3-7B	7B	Transformer	Mistral AI
DeepSeek-R1-8B	8B	Transformer	DeepSeek
Phi-4-14B	14B	Transformer	Microsoft
Gemma-2-27B	27B	Transformer	Google DeepMind
Qwen-2.5-72B	72B	Transformer	Alibaba Cloud

are based on Transformer decoder architectures, they differ in their pre-training corpora, parameter scale, and post-training alignment methods. In particular, some models employ additional optimization techniques, such as reinforcement learning-based techniques, to improve reasoning and instruction-following capabilities. These differences provide a diverse basis for evaluating Bengali text classification performance.

To ensure full transparency and reproducibility of our experiments, we utilized the official model weights hosted on the Hugging Face Hub. Specifically, the evaluated models correspond to the following repository identifiers:

`meta-llama/Llama-3.2-3B-Instruct`, `mistralai/Mistral-7B-Instruct-v0.3`, `deepseek-ai/DeepSeek-R1-Distill-Llama-8B`, `microsoft/phi-4`, `google/gemma-2-27b`, and `Qwen/Qwen2.5-72B-Instruct-AWQ`. Notably, for the Qwen-2.5-72B model, we employed the AWQ (Activation-aware Weight Quantization) version to fit the model within the available VRAM constraints while maintaining robust inference performance.

The overall specifications of the evaluated models are summarized in Table 3.

The six selected models for assessment follow the decoder-only Transformer paradigm introduced by Vaswani et al. [41] and later scaled successfully in GPT-style language models such as GPT-3 [42], where the self-attention mechanism is the central component.

The computation of the self-attention mechanism for the input sequence is as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

Here, Q denotes query, K denotes key, and V denotes value metrics, and the key-vector dimension is d_k . This is improved by a multi-head attention mechanism, through the computation of attention in diverse representational layers:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (2)$$

Here, the single head computes as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V). \quad (3)$$

Table 4 outlines the characteristics of six LLMs, including the architectural specification, the depth of layers, training parameters, and hidden dimensions.

LLaMA-3.2-3B

The LLaMA (Large Language Model Meta AI) series was developed by Meta [43]. It provides open and effective models for the use of multiple languages. In this series, the smallest version is the 3.2B version, and it is the most compact model with

Table 4. Detailed Architectural Specifications of Evaluated Large Language Models.

Model	Layers (L)	Attention Heads (h)	Hidden Dim (d_h)	Context Length	Vocab Size	Training Tokens
LLaMA-3.2-3B	28	28	3,072	8,192	128,256	15T
Mistral-V3-7B	32	32	4,096	32,768	32,000	7T
DeepSeek-R1-8B	32	32	4,096	4,096	128,256	2T
Phi-4-14B	40	32	5,120	16,384	32,064	1.5T
Gemma-2-27B	46	32	4,608	8,192	256,128	13T
Qwen-2.5-72B	80	64	8,192	131,072	151,936	18T

competence suitable for multiple languages, and for environments with limited resources. LLaMa-3.2-3B consists of a conventional transformer decoder structure with 28 layers. These individual layers include multi-head self-attention, which is accompanied by a feed-forward network.

A variation of multi-head attention named Grouped-Query Attention (GQA) [44] is implemented in this model. Grouped-Query Attention (GQA) preserves the multiple query heads, while reducing the number of key-value heads. The mathematical expression is as follows:

$$\text{GQA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (4)$$

While the concatenation resembles standard multi-head attention, GQA organizes the h query heads into g groups (where $g < h$). Each group shares a single key and value head. For LLaMA-3.2-3B, these queries consist of 28 query heads and 4 key-value bundles. This design decreases the memory requirements during the predictions, while ensuring the accuracy of the model, just like a full multi-head attention.

The SwiGLU activation function used in each layer of the feed-forward network is as follows:

$$\text{FFN}(x) = (\sigma(xW_1) \odot xW_2)W_3 \quad (5)$$

Here, $\sigma(x) = x \cdot \text{sigmoid}(x)$, where σ refers to the SiLU (also known as Swish) activation function, and $\odot$ refers to the element-wise multiplication. The dimension of intermediate feed-forward is $d_{ff} = 8192$, which is almost $2.67 \times d_h$.

LLaMA-3.2-3B was pre-trained on approximately 15 trillion text units from a huge multilingual text compilation, which covers more than 100 languages. The model utilizes the Byte-Pair-Encoding (BPE) tokenizer, which consists of 128,256 tokens maintaining the proper handling of complex scripts like Bengali. For positional encoding, the model uses Rotary Position Embeddings (RoPE), as follows:

$$\text{RoPE}(x_m, m) = \begin{bmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{bmatrix} \begin{bmatrix} x_m^{(1)} \\ x_m^{(2)} \end{bmatrix} \quad (6)$$

Here, m denotes the position index and θ denotes the frequency parameter. This encoding improves the model's capability to scale to longer sequences.

The overall parameter count is approximately as follows:

$$N_{\text{params}} = L(4d_h^2 + 2d_h d_{ff}) + Vd_h \approx 3.2 \times 10^9 \quad (7)$$

Here, L refers to 28 layers, $d_h = 3072$ refers to 3072 hidden dimensions, $d_{ff} = 8192$ refers to 8192 feed-forward dimensions, and V refers to 128,256 vocabulary size. Even though LLaMA-3.2-3B is on the smaller side, it still displays remarkable performance, making it ideal for zero-shot and few-shot multilingual evaluation.

Mistral-V3-7B

Mistral is a lightweight and effective large language model created by MistralAI; it contains 7 billion parameters [45]. The architecture is designed in such a way that it balances the prediction speed, reasoning capability, and computational efficiency. The original Mistral structure was improved with multiple enhancements to generate the third-generation model, Mistral-V3-7B. The most notable improvement was the Sliding Window Attention (SWA) mechanism, which helps with the effective processing of lengthy sequences.

The Sliding Window Attention (SWA) mechanism restricts each token’s attention computation to a specific window of size W :

$$\text{SWA}(Q, K, V)_i = \text{softmax}\left(\frac{Q_i K_{[i-W:i+W]}^T}{\sqrt{d_k}}\right) V_{[i-W:i+W]} \quad (8)$$

Here, the position index is i , and $[i - W : i + W]$ refers to the adjacent tokens that are near W . Mistral utilized $W(4096)$, maintains local coherence while enabling effective long-context modeling. This helps decrease the processing complexity from $O(n^2)$ to $O(n \cdot W)$, the n refers to the length of the sequence.

The structure is made up of 32 Transformer layers, a single layer contains 32 attention heads and 4096 hidden dimensions. The SwiGLU activation is implemented in the feed-forward network, where the intermediate dimension is as follows:

$$d_{ff} = 14,336 \approx 3.5 \times d_h \quad (9)$$

Rotary Positional Embeddings (RoPE) are configured for the coherence with the sliding window pattern. The tokenizer consists of 32,000 tokens, which are built on parts of sentences. Initially augmented for the European languages, with slight inclusion of South Asian languages.

Around 7 trillion tokens were used to pre-train the base model; these tokens were gathered from various sources such as academic texts, code, and web documents. [The model evaluated in this study corresponds to the instruction-tuned variant, Mistral-7B-Instruct-v0.3. Following the initial pre-training stage, this variant underwent additional instruction tuning and alignment procedures to improve instruction-following and response quality \[45\].](#) These post-training stages enhance the model’s ability to generate more accurate and contextually appropriate responses.

DeepSeek-R1-8B

DeepSeek AI introduced the DeepSeek-R1 family to significantly advance reasoning capabilities, particularly for complex tasks involving mathematics, coding, and STEM problem-solving [46]. [The specific model evaluated in this study is officially named DeepSeek-R1-Distill-Llama-8B, an 8-billion-parameter distilled model built upon the Llama-3.1-8B architecture. This variant inherits advanced reasoning capabilities distilled from the larger DeepSeek-R1 model while maintaining a significantly smaller computational footprint.](#)

[The architecture of DeepSeek-R1-Distill-Llama-8B follows a standard dense Transformer decoder design consisting of 32 attention heads, 32 hidden layers, and a hidden dimension of 4096. Therefore, reinforcement learning is not considered an architectural component, but rather a post-training optimization and alignment technique used to improve reasoning performance.](#) The model employs self-attention mechanisms, feed-forward networks, Rotary Positional Embeddings (RoPE), and RMSNorm for efficient language modeling.

The model is initially trained using causal language modeling through next-token prediction:

$$\mathcal{L}_{\text{LM}} = - \sum_{t=1}^T \log P(x_t | x_{<t}; \theta) \quad (10)$$

where x_t denotes the token at step t , $x_{<t}$ represents the preceding tokens, and θ denotes the model parameters.

The full DeepSeek-R1 training pipeline incorporates reinforcement learning, specifically Group Relative Policy Optimization (GRPO), to enhance reasoning abilities. The distilled model evaluated in this study inherits reasoning patterns learned through this process. The optimization objective is expressed as:

$$\mathcal{L}_{\text{RL}} = -\mathbb{E}_{y \sim \pi_{\theta}(y|x)} [r(x, y) - \beta \text{KL}(\pi_{\theta}(y|x) \parallel \pi_{\text{ref}}(y|x))] \quad (11)$$

where $r(x, y)$ denotes the reward assigned to output y for input x , π_{θ} is the policy being optimized, π_{ref} is the reference policy, and β controls the divergence penalty. The reward mechanism consists of accuracy rewards and format rewards, encouraging both correct answers and structured reasoning outputs.

The model employs Rotary Positional Embeddings (RoPE) for positional encoding and RMSNorm for normalization:

$$\text{RMSNorm}(x) = \frac{x}{\text{RMS}(x)} \cdot \gamma, \quad \text{RMS}(x) = \sqrt{\frac{1}{d} \sum_{i=1}^d x_i^2} \quad (12)$$

where γ is a trainable scaling parameter and d denotes the dimensionality of the hidden representation.

Phi-4-14B

The Phi models are generated by Microsoft to focus on the quality of data and compilation, along with scope of data [47]. The Phi-4-14B models are a combination of size and computational efficacy. Additionally, this model presents noteworthy outcomes in structured identification and reasoning problems. The structure of Phi-4-14B consists of a conventional Transformer design, which includes 40 layers, 32 attention heads and 5120 hidden dimensions.

The principal alteration in Phi-4 is found in its data processing method rather than the architecture of the model. The Phi-4-14B model is trained with the intention of utilizing the basic causal language modeling:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \sum_{t=1}^{T_i} \log P_{\theta}(x_t^{(i)} | x_{<t}^{(i)}) \quad (13)$$

Here, N denotes the serial of the training sequence and T_i refers to the length of a single sequence i .

The training data collection of Phi-4-14B consists of approximately 1.5 trillion tokens, all these tokens undergoes detailed assessment and filtering. Single record obtains a composite quality score $Q(d)$, which is calculated as follows:

$$Q(d) = w_1 q_{\text{edu}}(d) + w_2 q_{\text{fact}}(d) + w_3 q_{\text{diverse}}(d) - w_4 q_{\text{toxic}}(d) \quad (14)$$

In the equation, q_{edu} represents educational worth, q_{fact} represents factual accuracy, q_{diverse} measures lexical diversity, and q_{toxic} represents unfavorable or biased

content. Individual records that satisfy the threshold condition $Q(d) > \tau$ are included in the training dataset.

Furthermore, Phi-4-14B amalgamate high-quality artificial data created by bigger models and verified by human specialists. The procedure is shown as follows:

$$d_{\text{syn}} = \text{LLM}_{\text{large}}(p_{\text{template}}, k_{\text{knowledge}}) \rightarrow \text{Human}_{\text{verify}} \quad (15)$$

Here, p_{template} refers to the predefined prompt pattern, and $k_{\text{knowledge}}$ refers to compiled logical information. Human authentication maintains the logical uniformity and reliability of all the organized data.

The tokenizer used by Phi-4-14B consists of 32,064 tokens, and enables a context length of about 16,384 tokens. The proportions of the feed-forward network are as follows:

$$d_{ff} = 4 \times d_h = 20,480 \quad (16)$$

The estimated attribute count is given by the following:

$$N_{\text{params}} = L \times (4d_h^2 + 2d_h d_{ff}) + V \times d_h \approx 14 \times 10^9 \quad (17)$$

Gemma-2-27B

Gemma-2 is an advanced open-weights large language model developed by Google DeepMind [48]. The 27-billion-parameter variant was designed to deliver strong performance and inference efficiency through several architectural improvements over the standard Transformer framework.

The model architecture consists of 46 Transformer layers with a hidden dimension of 4608. To improve inference efficiency and reduce memory requirements, Gemma-2-27B employs Grouped-Query Attention (GQA) with 32 attention heads and 16 key-value heads [48]. The attention mechanism can be expressed as:

$$\text{GQA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O \quad (18)$$

where multiple query heads share key-value representations, reducing KV-cache memory during autoregressive generation.

Gemma-2 further introduces an interleaved attention mechanism that alternates between global attention and local Sliding Window Attention (SWA) layers. This design enables efficient modeling of both long-range dependencies and local contextual information.

The model employs Rotary Positional Embeddings (RoPE) for positional encoding [48]. In addition, Gemma-2 applies logit soft-capping to prevent excessively large logits during training and inference, thereby improving numerical stability.

Gemma-2-27B was trained on approximately 13 trillion tokens collected from web documents, source code, and scientific articles [48]. The model uses a SentencePiece tokenizer with a vocabulary size of 256,128 tokens, supporting a broad range of languages and scripts. This large multilingual vocabulary enables effective tokenization across diverse linguistic domains and writing systems.

Qwen-2.5-72B

The largest model evaluated in this study is Qwen-2.5-72B-Instruct, an instruction-tuned large language model developed by Alibaba Cloud [49]. The model contains approximately 72 billion parameters and was trained on a large-scale multilingual corpus comprising approximately 18 trillion tokens. Qwen-2.5 supports

more than 29 languages and demonstrates strong performance in reasoning, coding, mathematics, and multilingual understanding.

The architecture consists of 80 Transformer layers with a hidden dimension of 8192. To improve inference efficiency and reduce memory consumption, Qwen-2.5-72B employs Grouped-Query Attention (GQA) with 64 query attention heads and 8 key-value heads [44]. The head configuration is given by:

$$h_{\text{query}} = 64, \quad h_{\text{kv}} = 8, \quad \text{ratio} = \frac{h_{\text{query}}}{h_{\text{kv}}} = 8 \quad (19)$$

The attention computation for a single group can be expressed as:

$$\text{head}_{i,g} = \text{Attention}(Q_i W_i^Q, K_g W_g^K, V_g W_g^V) \quad (20)$$

where i denotes the query head index and g identifies the corresponding shared key-value group.

Qwen-2.5 further incorporates Rotary Positional Embeddings (RoPE), SwiGLU activation functions, RMSNorm, and Attention QKV bias to improve representation learning and inference stability. The model supports long-context processing and can handle sequences substantially longer than conventional Transformer-based LLMs.

The model utilizes a byte-level BPE tokenizer with a vocabulary size of 151,936 tokens [49]. This large vocabulary facilitates effective tokenization across diverse languages and writing systems, including Bengali. To accommodate the large parameter count within available hardware constraints, the AWQ (Activation-aware Weight Quantization) version of Qwen-2.5-72B was used in this study while preserving strong inference performance.

Prompting Strategies

To improve methodological transparency, we explicitly describe the prompt design procedure adopted in this study. All prompt instructions used in both zero-shot and few-shot settings were written in English, while the input texts and few-shot examples were presented in Bengali. English instructions were selected because the evaluated open-source LLMs are primarily instruction-tuned using English prompts, thereby providing a consistent and model-agnostic evaluation framework.

To analyze the in-context learning competence of the applied Large Language Models (LLMs), two prompting techniques were implemented during the experiment.

Zero-shot prompting: For the zero-shot approach, a task instruction prompt is given to the model, for example detect the sentiment or emotion. Additionally, no more samples or context are provided with the prompt. Following this, models utilize their general knowledge initiated completely from the provided prompt to create responses for task completion.

Few-shot prompting (five-shot): For the few-shot approach, the model is not provided with any instructions; instead, it is given five examples along with the outputs of the examples. Then the model proceeds to generate responses by deducing the task criteria from the five examples. This enhances the accuracy of models in solving the tasks.

The structure of these techniques adheres to the previous papers on in-context learning [8, 12], because previously, after giving the models a few samples, the models performed better and adjusted to the new data more, specifically in the case of low-resource language situations.

To ensure a fair comparison, the same prompt template structure was applied across all evaluated models within each task. For few-shot prompting, five labeled examples were selected from the corresponding training dataset and remained identical

across all models. A fixed random seed (seed = 42) was used throughout the experiments to ensure reproducibility. No model-specific prompt engineering, optimization, or calibration was performed. This design ensured that performance differences reflected model capability rather than variations in prompt construction.

Task-related prompts were generated for Bengali text classification, the four types of classifications include *Sentiment Analysis*, *Emotion Recognition*, *Hate Speech Detection*, and *Fake News Detection*. The prompt was drafted individually to properly describe the task, explain the label space, and stipulate the necessary JSON output pattern for uniform parsing and assessment.

This consistent pattern enables equal comparison among the LLMs. We differentiate the zero-shot and few-shot systems to analyze if the incorporation of contextual samples enhanced the classification performance and accuracy for Bengali text classification.

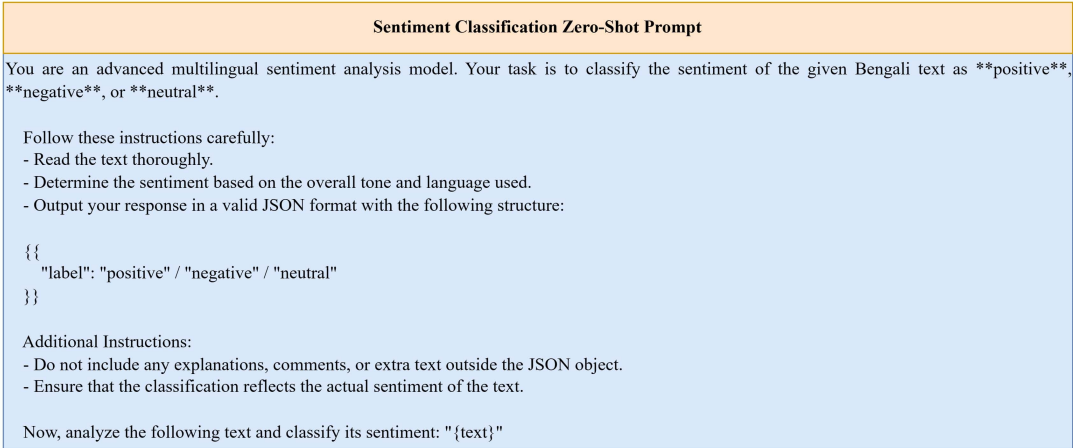


Fig 2. Prompt used for zero-shot Bengali sentiment analysis.

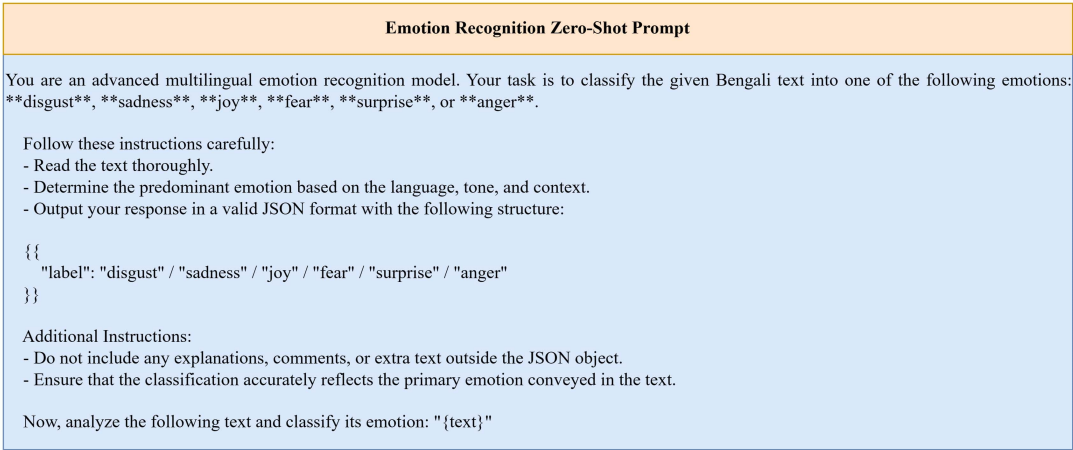
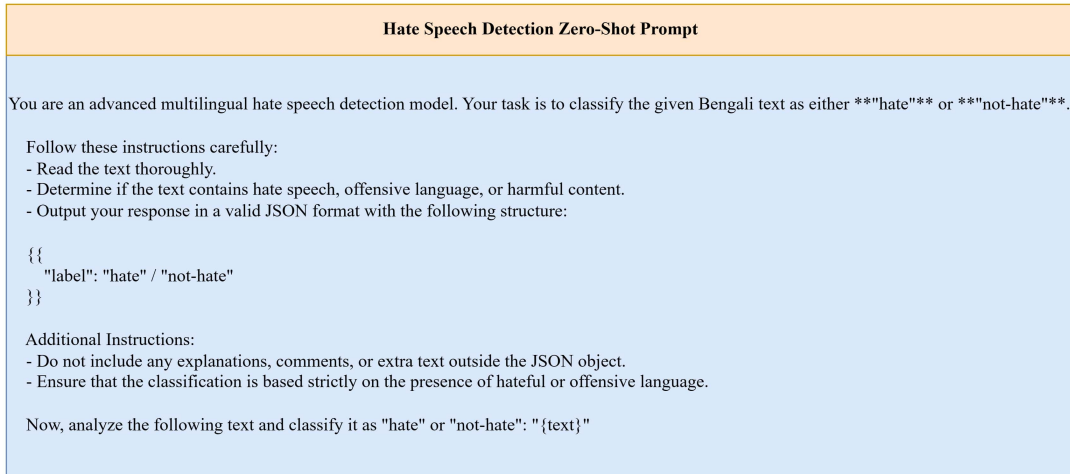
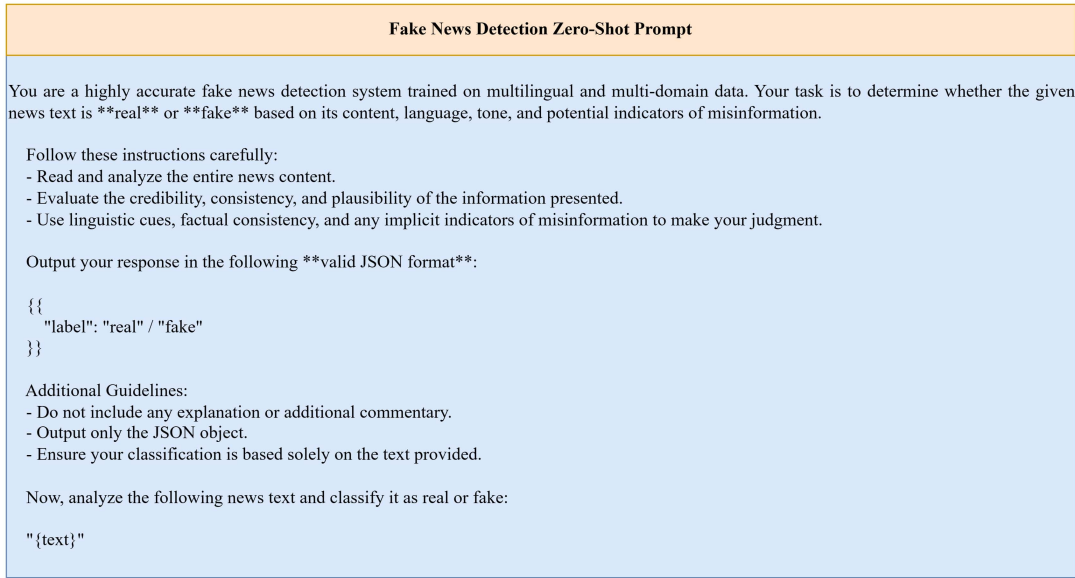


Fig 3. Prompt used for zero-shot Bengali emotion recognition.



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Fig 4. Prompt used for zero-shot Bengali hate speech detection.



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Fig 5. Prompt used for zero-shot Bengali fake news detection.

Sentiment Classification Few-Shot Prompt
You are an advanced multilingual sentiment analysis model. Your task is to classify Bengali text as having positive, negative, or neutral sentiment. Below are 5 example texts with their correct sentiment labels: Example 1: Text: "আজকের দিনটা দারুণ কাটলো, খুব ভালো লাগছে।" Label: "positive" Example 2: Text: "এই রাতটা একেবারেই খারাপ, চলাফেরা করাই মুশকিল।" Label: "negative" Example 3: Text: "আজ সকাল থেকে বৃষ্টি হচ্ছে।" Label: "neutral" Example 4: Text: "ওর সঙ্গে দেখা হয়ে খুব ভালো লাগলো।" Label: "positive" Example 5: Text: "ট্রেনটা আজ আবার দেরিতে এসেছে, বিরক্ত লাগছে।" Label: "negative" Now, classify the sentiment of the following text: Text: "{text}" Output your response in a valid JSON format with the following structure: <pre>{{ "label": "positive" / "negative" / "neutral" }}</pre> Instructions: - Output only the JSON object. - Do not include explanations or comments. - Make sure the sentiment reflects the actual tone and context of the text.

Fig 6. Prompt used for few-shot Bengali sentiment analysis.

Emotion Recognition Few-Shot Prompt
You are an advanced multilingual emotion recognition model. Your task is to classify Bengali text into one of the following emotions: **disgust**, **sadness**, **joy**, **fear**, **surprise**, or **anger**. Below are 5 example texts with their correct emotion labels: Example 1: Text: "ওর এমন ব্যবহারে আমি ভীষণ ঘৃণা বোধ করেছি।" Label: "disgust" Example 2: Text: "আজ আমার বাবার মৃত্যুবার্ষিকী, খুব কষ্ট পাচ্ছি।" Label: "sadness" Example 3: Text: "আমি পরীক্ষায় প্রথম হয়েছি! খুব খুশি লাগছে।" Label: "joy" Example 4: Text: "রাতে হঠাৎ দরজার আওয়াজ পেয়ে খুব ভয় লাগলো।" Label: "fear" Example 5: Text: "ও এমন কথা বলবে ভাবতেই পারিনি!" Label: "surprise" Now, classify the following text into one of the predefined emotions. Text: "{text}" Output your response in a valid JSON format with the following structure: <pre>{{ "label": "disgust" / "sadness" / "joy" / "fear" / "surprise" / "anger" }}</pre> Instructions: - Output only the JSON object. - Do not include explanations or comments. - Ensure the label reflects the main emotion conveyed in the text.

Fig 7. Prompt used for few-shot Bengali emotion recognition.

Hate Speech Detection Few-Shot Prompt
You are an advanced multilingual hate speech detection model. Your task is to classify Bengali text as either ***"hate"*** or ***"not-hate"***. Below are 5 example texts with their correct hate speech labels: Example 1: Text: "এই মানুষগুলো সব কুচক্রি, এদের দেশ থেকে তাড়ানো উচিত।" Label: "hate" Example 2: Text: "আজকে অনেক গরম পড়েছে, বাইরে যাওয়া কঠিন।" Label: "not-hate" Example 3: Text: "ওরা একেবারে অকর্মা, সমাজের বোঝা মাত্র।" Label: "hate" Example 4: Text: "তোমার বক্তব্যটা শুনে ভালো লাগলো, অনেক কিছু শিখলাম।" Label: "not-hate" Example 5: Text: "এমন লোকদের দেখলেই রাগ ধরে, এদের পেটাতে ইচ্ছে করে।" Label: "hate" Now, analyze the following text and classify it as "hate" or "not-hate": Text: "{text}" Output your response in a valid JSON format with the following structure: <pre>{{ "label": "hate" / "not-hate" }}</pre> Instructions: - Output only the JSON object. - Do not include explanations or comments. - Ensure that the classification is based strictly on the presence of hateful, offensive, or harmful language.

Fig 8. Prompt used for few-shot Bengali hate speech detection.

Fake News Detection Few-Shot Prompt
You are a highly accurate fake news detection system trained on multilingual and multi-domain data. Your task is to determine whether the given news text is **real** or **fake** based on its content, language, tone, and potential indicators of misinformation. Below are 5 example news texts along with their correct classifications: Example 1: Text: "সরকার ঘোষণা করেছে আগামীকাল থেকে দেশের সব শিক্ষাপ্রতিষ্ঠান পুনরায় খোলা হবে।" Label: "real" Example 2: Text: "একটি গবেষণায় দেখা গেছে যে প্রতিদিন লবঙ্গ খেলে করোনা সম্পূর্ণরূপে নিরাময় হয়।" Label: "fake" Example 3: Text: "চট্টগ্রামে ভূমিকম্পে ক্ষতিগ্রস্ত হয়েছে বহু বাড়িঘর, উদ্ধার কাজ চলছে।" Label: "real" Example 4: Text: "চাঁদের বৃকে বিশালাকার প্রাণী দেখা গেছে, নাসা নিশ্চিত করেছে।" Label: "fake" Example 5: Text: "বাংলাদেশ ক্রিকেট দল আগামী মাসে ভারত সফরে যাচ্ছে বলে জানিয়েছে বিসিবি।" Label: "real" Now, analyze the following news text and classify it as **real** or **fake**: Text: "{text}" Respond in the following **valid JSON format**: <pre> {{ "label": "real" / "fake" }}</pre> Instructions: - Do not include any explanation or commentary. - Output only the JSON object. - Ensure your classification is based solely on the content and plausibility of the news text.

Fig 9. Prompt used for few-shot Bengali fake news detection.

Experimental Setup

All model evaluations were conducted using the Hugging Face Transformers framework and the *vLLM* inference engine, which provides efficient memory management and scalable deployment for large language models. Two computing environments were used during this study. A local workstation equipped with an AMD Ryzen 5 8400F processor (4.20 GHz, 6-core), 16 GB RAM, and an NVIDIA GeForce RTX 4060 GPU (8 GB VRAM) was employed for prompt development, preliminary testing, and evaluation of smaller models.

For large-scale inference involving Qwen-2.5-72B-Instruct-AWQ, a cloud-based NVIDIA RTX PRO 6000 GPU with approximately 96 GB VRAM was utilized through the RunPod cloud platform. The model was deployed in its AWQ-quantized form to improve memory efficiency while maintaining predictive performance. This cloud-based configuration was necessary because the memory requirements of the 72B-parameter model exceed the capacity of an 8 GB consumer-grade GPU. No multi-GPU cluster was used. All Qwen-2.5-72B-Instruct-AWQ inference experiments were conducted on a single cloud-based NVIDIA RTX PRO 6000 GPU through the RunPod platform.

To ensure a fair comparison, all models were evaluated using a unified deterministic decoding configuration. Generation was performed using greedy decoding (`do_sample=False`) with a maximum output length of 512 tokens. A fixed random

Table 5. Inference Configuration Parameters.

Parameter	Value
Random Seed	42
Decoding Strategy	Greedy Decoding (<code>do_sample=False</code>)
Max Tokens	512
Cloud GPU	NVIDIA RTX PRO 6000 (96 GB VRAM)
Cloud Platform	RunPod
Local Development GPU	NVIDIA GeForce RTX 4060 (8 GB VRAM)
Processor	AMD Ryzen 5 8400F (4.20 GHz, 6-Core)
System RAM	16 GB
Quantization	AWQ (4-bit)
Inference Engine	vLLM

seed of 42 was used throughout the experiments to ensure reproducibility. The same decoding configuration was applied across all datasets and prompting strategies so that performance differences primarily reflected model capability rather than variations in generation settings.

All experiments were executed on 64-bit operating systems with CUDA and PyTorch properly configured. The combination of vLLM, AWQ quantization, and cloud-based GPU resources provided through RunPod enabled efficient and reproducible evaluation of all models, including Qwen-2.5-72B-Instruct-AWQ.

Evaluation Metrics

The models were evaluated using established evaluation metrics: *Accuracy*, *Precision*, *Recall*, and *Macro-averaged F1-score*.

Precision and Recall demonstrate the relationship between false positives and false negatives. Given the presence of class imbalance in some of our datasets, the Macro-averaged F1-score was selected as the primary robust metric for evaluation, as recommended for datasets where both classes matter equally. Unlike Accuracy, which may provide misleadingly high values in imbalanced scenarios by favoring majority-class predictions, the Macro-averaged F1-score computes the F1-score independently for each class and then averages them. This ensures that both majority and minority classes contribute equally to the final score, providing a more balanced assessment of classification performance.

While Accuracy provides the overall rate of correct predictions [50], it is reported alongside the Macro-averaged F1-score rather than serving as the sole primary metric due to its limitations in imbalanced classification settings.

Moreover, a *confusion matrix* was generated for each model to obtain True Positives (TP), False Positives (FP), False Negatives (FN), and True Negatives (TN), providing additional insight into model-specific error patterns and class-wise performance.

The formal definitions of the evaluation metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (21)$$

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \quad (22)$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \quad (23)$$

$$\text{F1-score}_c = \frac{2 \times \text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \quad (24)$$

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \text{F1-score}_c \quad (25)$$

Here, c denotes an individual class, and C represents the total number of classes.

- **Accuracy** calculates the overall performance by taking the ratio of correct predictions to the total number of predictions.
- **Precision** measures the proportion of correctly predicted positive instances among all positive predictions made by the model.
- **Recall** assesses the proportion of actual positive instances that were correctly identified by the model.
- **Macro F1-score** computes the F1-score for each label and finds their unweighted mean. This treats all classes equally, making it highly effective for evaluating imbalanced datasets where minority classes are as important as majority classes.

Statistical Significance Testing

To assess whether the observed performance differences between the evaluated Large Language Models (LLMs) were statistically meaningful, we performed paired permutation tests on the model predictions. Because all models were evaluated on the same test instances, a paired testing framework was adopted to account for the dependency between predictions and to provide a more reliable comparison.

For each dataset and prompting setting, the observed differences in Accuracy and Macro F1-score between the compared models were first calculated. A permutation-based resampling procedure was then applied by randomly exchanging the predictions of the two models for individual test samples while preserving the ground-truth labels. This process was repeated 1,000 times to generate a null distribution representing the expected performance differences under the assumption that both models performed equivalently.

The p-value was estimated as the proportion of permutations that produced a performance difference greater than or equal to the observed difference. Statistical significance was assessed using a significance level of $\alpha = 0.05$. A p-value below 0.05 indicates that the observed difference is unlikely to have occurred by chance and can therefore be considered statistically significant.

A permutation-based approach was selected because it is non-parametric and does not require assumptions regarding the underlying distribution of evaluation metrics. Consequently, it is well suited for comparing classification models on shared benchmark datasets. In this study, the test was applied to compare the prediction outputs of Qwen-2.5-72B and Gemma-2-27B, the two strongest-performing models across the evaluated tasks.

Confidence Interval Estimation

To quantify the uncertainty associated with model performance estimates, 95% confidence intervals (CIs) were computed for the Macro F1-score using bootstrap resampling. For each model and evaluation setting, 1,000 bootstrap samples were generated by randomly sampling the test instances with replacement. The Macro

Table 6. Paired permutation test results comparing Qwen-2.5-72B and Gemma-2-27B.

Dataset	Setting	Accuracy p-value	Macro F1 p-value
Emotion	Zero-shot	1.000	0.943
Emotion	Few-shot	0.003	0.016
Fake News	Zero-shot	0.475	0.306
Fake News	Few-shot	0.876	0.658
Hate Speech	Zero-shot	0.007	0.004
Hate Speech	Few-shot	0.001	<0.001
Sentiment	Zero-shot	<0.001	0.101
Sentiment	Few-shot	0.023	0.055

F1-score was calculated for each bootstrap sample, and the 2.5th and 97.5th percentiles of the resulting distribution were used to construct the 95% confidence interval. This non-parametric approach does not assume any specific distribution of the evaluation metric and is widely used for estimating uncertainty in machine learning evaluation studies.

Results and Analysis

This section presents the performance of various LLMs on two prompting settings across four classification tasks. The associated results for each prompting strategy are described as follows.

Statistical Significance Analysis

To determine whether the observed performance differences between the evaluated LLMs were statistically significant, paired permutation tests were conducted using 1,000 random permutations. The analysis focused on the two strongest-performing models, Qwen-2.5-72B and Gemma-2-27B, across all datasets and prompting settings. Table 6 presents the corresponding p-values for accuracy and macro F1-score.

The results reveal that the statistical significance of the observed performance differences varies across tasks. For hate-speech detection, significant differences were observed under both zero-shot and few-shot settings for accuracy and macro F1-score ($p < 0.05$), indicating a consistent advantage of one model over the other. Similarly, emotion recognition under the few-shot setting exhibited statistically significant differences for both evaluation metrics.

In contrast, no statistically significant differences were observed for fake-news detection in either prompting setting ($p > 0.05$), suggesting that Qwen-2.5-72B and Gemma-2-27B achieved comparable performance on this task. For sentiment analysis, the findings were mixed. Although the accuracy differences were statistically significant in both prompting settings, the corresponding macro F1-score differences did not consistently reach the significance threshold.

Overall, these results suggest that numerical differences in performance metrics should be interpreted with caution, as they do not always correspond to statistically significant performance gains. Therefore, both evaluation metrics and statistical significance tests should be considered when comparing LLM performance.

Table 7. 95% bootstrap confidence intervals for Macro F1-score.

Dataset (Test Size)	Prompt Setting	Qwen-2.5-72B F1 (95% CI)	Gemma-2-27B F1 (95% CI)
Emotion (625)	Zero-shot	61.91 (58.08, 65.96)	61.80 (57.95, 65.39)
Emotion (625)	Few-shot	66.80 (63.11, 70.17)	63.01 (59.59, 66.37)
Fake News (508)	Zero-shot	82.91 (79.57, 86.11)	84.33 (80.87, 87.37)
Fake News (508)	Few-shot	83.68 (80.28, 86.80)	84.17 (80.98, 87.34)
Hate Speech (1000)	Zero-shot	79.88 (77.49, 82.28)	76.11 (73.38, 78.71)
Hate Speech (1000)	Few-shot	78.25 (75.65, 80.67)	73.39 (70.56, 76.04)
Sentiment (1586)	Zero-shot	56.84 (54.45, 59.22)	58.66 (56.38, 61.25)
Sentiment (1586)	Few-shot	58.82 (56.45, 61.11)	60.68 (58.37, 63.18)

Confidence Interval Analysis

To account for the varying test-set sizes across the evaluated tasks, 95% bootstrap confidence intervals were computed for the Macro F1-score of the two strongest-performing models, Qwen-2.5-72B and Gemma-2-27B. Table 7 reports the corresponding confidence intervals.

As expected, datasets with smaller test sizes, such as Emotion Recognition and Fake News Detection, exhibited relatively wider confidence intervals than larger datasets. In contrast, the Sentiment Analysis dataset, which contains the largest test set, produced comparatively narrower intervals. Overall, the confidence intervals provide additional evidence regarding the stability and reliability of the reported model performance estimates.

Zero-Shot Performance

Table 8 presents the zero-shot performance of all evaluated large language models (LLMs) across the four Bengali text classification tasks. For the sentiment Analysis task, Gemma-2-27B reported the overall best results on accuracy (62.04%), recall (62.04%), and **Macro-F1** score (58.66%) a word-based approach that generalizes very well even without in-context usage of examples. In comparison, Qwen-2.5-72B achieved the best precision with 66.12%, meaning it was better at suppressing false positives and generating confident predictions.

For the Emotion Recognition benchmark, we compare a combination of Qwen-2.5-72B and Gemma-2-27B exhibited comparable results consistently, with an accuracy of 61.76%. Their **Macro-F1** scores of 66.80% and 63.01%, respectively, indicate that such large multilingual models have a reasonable ability to recognise emotion expressions in Bengali text without task-specific fine-tuning.

Among the Hate Speech Detection systems, Qwen-2.5-72B achieved the best results in terms of overall performance, with an accuracy of 79.90% and a **Macro-F1** score of 79.88%, demonstrating that this model is very robust in detecting offensive or harmful material. We achieved the best accuracy of 84.45% and a **Macro-F1** score of 84.33% in Fake News Detection on Gemma-2-27B, indicating it is competent to separate fact from fake.

In general, the results suggest that the larger models, especially Qwen-2.5-72B and Gemma-2-27B perform better than the smaller models in all tasks. The success of these models is due to their larger number of parameters, multilingual training, and wider token coverage, contributing to a more comprehensive understanding of Bengali syntactic and semantic context. Smaller models such as LLaMA-3.2-3B, Mistral-V3-7B, and DeepSeek-R1-8B were less competitive. This could be partially because they have a smaller capacity and receive less Bengali exposure in pretraining, so that their generalisation is weaker for low-resource language tasks.

Table 8. Zero-shot evaluation results of Large Language Models (LLMs) on various Bengali text classification tasks.

Model	Sentiment Analysis				Emotion Recognition			
	Acc.	P.	R.	Macro-F1	Acc.	P.	R.	Macro-F1
LLaMA-3.2-3B	52.21	54.62	52.21	49.62	46.24	49.08	46.24	46.35
Mistral-V3-7B	51.32	54.49	51.32	50.20	39.86	39.40	36.96	45.18
DeepSeek-R1-8B	50.82	55.15	50.82	49.90	45.92	47.48	45.92	45.85
Phi-4-14B	59.84	61.88	59.84	57.73	59.76	59.73	59.76	57.90
Gemma-2-27B	62.04	65.12	62.04	58.66	61.76	68.47	61.76	63.01
Qwen-2.5-72B	58.07	66.12	58.07	56.84	61.76	67.67	61.76	66.80
Model	Hate Speech Detection				Fake News Detection			
	Acc.	P.	R.	Macro-F1	Acc.	P.	R.	Macro-F1
LLaMA-3.2-3B	57.10	59.49	57.10	54.22	77.76	77.95	77.76	77.72
Mistral-V3-7B	64.70	65.58	64.70	64.19	61.11	76.02	61.11	55.45
DeepSeek-R1-8B	59.30	63.97	59.30	56.40	74.30	77.36	74.30	72.39
Phi-4-14B	72.50	73.52	72.50	72.17	77.36	79.64	77.36	76.91
Gemma-2-27B	76.50	78.33	76.50	76.11	84.45	85.57	84.45	84.33
Qwen-2.5-72B	79.90	80.05	79.90	79.88	83.27	86.26	83.27	82.91

Few-Shot Performance

Table 9 presents the performance of the evaluated large language models (LLMs) in the few-shot setting, where each model was provided with five labelled examples per task. Overall, Qwen-2.5-72B performs the best in the Emotion Recognition and Hate Speech Detection tasks, with a Macro-F1 score of 78.25% and 78.25%, respectively. Gemma-2-27B achieves outstanding results in the Sentiment Analysis and Fake News Detection tasks with a Macro-F1 score of 60.68% and 84.17%. These results suggest that Qwen-2.5-72B and Gemma-2-27B consistently perform well across different domains and utilize few-shot context effectively.

Compared with the zero-shot setting, the effect of few-shot prompting is task-dependent and varies across models. Noticeable improvements are observed in several cases, particularly for Emotion Recognition, where most models achieve higher Macro-F1 scores after receiving in-context examples. This suggests that modest supervision can help models better capture nuanced linguistic and contextual cues in Bengali. However, the benefits are not universal. For some tasks, such as Fake News Detection, performance remains largely unchanged or decreases slightly for certain models. Similarly, Qwen-2.5-72B records a small reduction in Hate Speech Detection Macro-F1 score (79.88% to 78.25%) compared with its zero-shot performance. These findings indicate that when a model already possesses strong task-specific capabilities from pre-training, additional in-context examples may provide limited benefit and can occasionally lead to marginal performance degradation.

Both Mistral-V3-7B and DeepSeek-R1-8B show relatively weak performance compared with the other models. For example, their Macro-F1 scores in Sentiment Analysis are 48.09% and 49.47%, while in Emotion Recognition they achieve 63.07% and 61.47%. This inferior performance can be explained partly because of the smaller parameter size and less multilingual training, as well as lesser exposure to Bengali in

Table 9. Few-shot evaluation results of Large Language Models (LLMs) on various Bengali text classification tasks.

Model	Sentiment Analysis				Emotion Recognition			
	Acc.	P.	R.	Macro-F1	Acc.	P.	R.	Macro-F1
LLaMA-3.2-3B	57.12	57.50	57.12	54.61	46.72	63.35	46.72	62.85
Mistral-V3-7B	47.92	58.03	47.92	48.09	46.24	57.79	46.24	63.07
DeepSeek-R1-8B	50.63	54.20	50.63	49.47	46.56	45.69	46.56	61.47
Phi-4-14B	58.32	64.24	58.32	56.44	56.00	64.95	56.00	70.56
Gemma-2-27B	62.55	67.93	62.55	60.68	61.12	71.66	61.12	73.39
Qwen-2.5-72B	60.28	66.86	60.28	58.82	65.76	70.79	65.76	78.25
Model	Hate Speech Detection				Fake News Detection			
	Acc.	P.	R.	Macro-F1	Acc.	P.	R.	Macro-F1
LLaMA-3.2-3B	64.40	67.28	64.40	62.85	69.29	80.05	69.29	66.27
Mistral-V3-7B	63.10	63.14	63.10	63.07	63.58	74.84	63.58	58.93
DeepSeek-R1-8B	62.30	63.46	62.30	61.47	67.32	71.69	67.32	65.59
Phi-4-14B	71.00	72.33	71.00	70.56	81.69	83.25	81.69	81.48
Gemma-2-27B	74.30	78.15	74.30	73.39	84.45	87.03	84.45	84.17
Qwen-2.5-72B	78.40	79.18	78.40	78.25	84.06	87.50	84.06	83.68

the pre-training. Larger multilingual models such as Qwen-2.5-72B and Gemma-2-27B exhibit more robust generalization ability, verifying the importance of scale and diverse multilingual data in few-shot learning for low-resource languages.

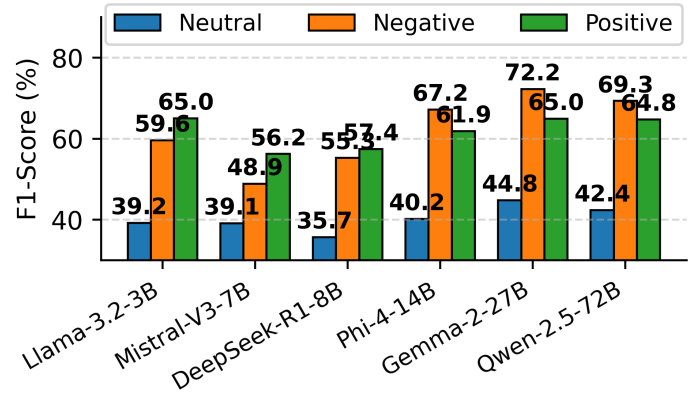
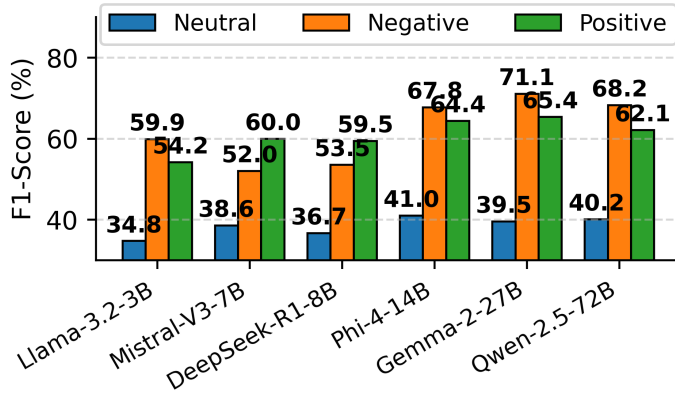
Class-Wise Performance Analysis

This section discusses the LLMs’ performance across different classes of the four classification tasks.

LLMs Performance Across Sentiment Categories

Fig.10 presents the F1-scores of each sentiment class across all large language models (LLMs) for both zero-shot and few-shot prompting settings. In the zero-shot condition, all models achieve higher F1 scores on the negative and positive sentiment classes when comparing with Gemma-2-27B model and Qwen-2.5-72B with the most optimal performance. However, the neutral class remains inferior in F1-score across all models, consistently showing that separating neutral sentiment is still difficult in the absence of context examples. This difficulty is partially exacerbated by the fact that the neutral class is underrepresented in the dataset (361 samples compared to 654 positive and 571 negative).

Fig.10 presents the F1-scores of each sentiment class across all large language models (LLMs) for both zero-shot and few-shot prompting settings. In the zero-shot condition, all models achieve higher F1 scores on the negative and positive sentiment classes when comparing with Gemma-2-27B model and Qwen-2.5-72B with the most optimal performance. However, the neutral class remains inferior in F1-score across all models, consistently showing that separating neutral sentiment is still difficult in the absence of context examples. This difficulty is partially exacerbated by the fact that



(a) Zero-shot setting

(b) Few-shot setting

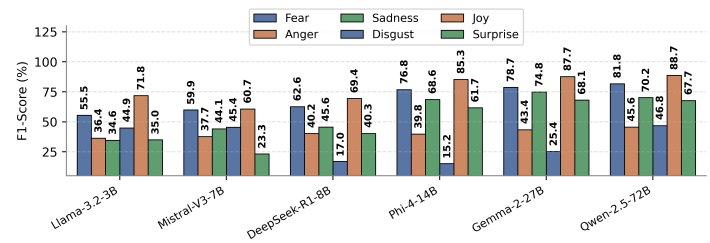
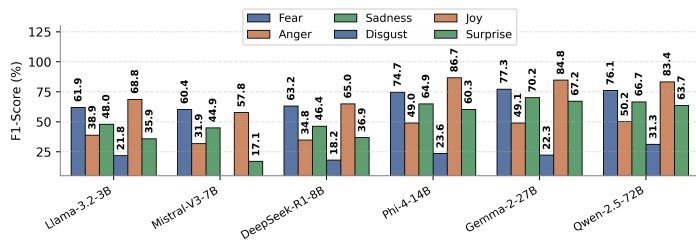
Fig 10. Class-wise sentiment classification performance for two prompt settings across LLMs.

the neutral class is underrepresented in the dataset (361 samples compared to 654 positive and 571 negative). This difficulty is partially exacerbated by the fact that the neutral class is underrepresented in the dataset (361 samples compared to 654 positive and 571 negative). This difficulty is partially exacerbated by the fact that the neutral class is underrepresented in the dataset (361 samples compared to 654 positive and 571 negative).

The trend is similar in the few-shot scenario, with substantial improvements for the neutral class observed across several models (including DeepSeek-R1-8B, which struggles with neutral sentiment under zero-shot conditions but benefits from few-shot examples). For example, Gemma-2-27B shows an increase of approximately five percentage points in the neutral category when moving from zero-shot to few-shot prompting. This upside reveals that few-shot prompting enables models to better understand the nuanced linguistic embedding of and context cues in the input text. Because of being exposed to these labelled examples, the models capture ambiguous or mixed sentiment expression much better, and achieve a more balanced performance across sentiment categories.

LLMs Performance Across Emotion Categories

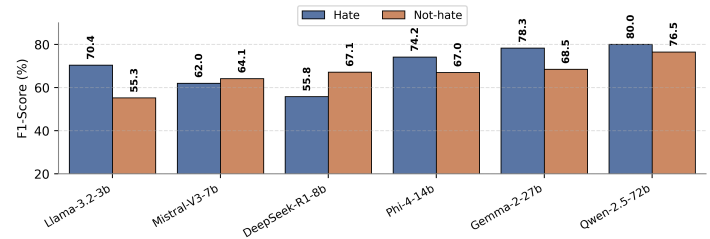
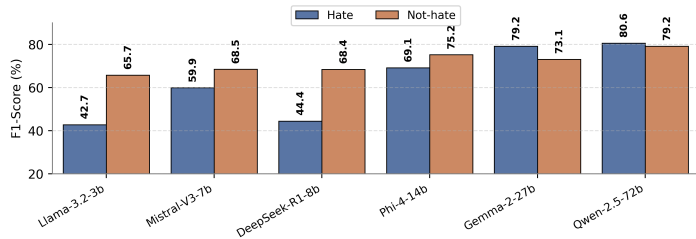
Fig.11 shows the F1 scores across emotion categories. As we can see in the zero-shot setting, the performance of LLMs is highly heterogeneous on emotion types, from Joy and Fear classes with higher F1 scores to Disgust and Anger categories being more difficult to classify accurately. Qwen-2.5-72B and Gemma-2-27B again show strong performance, except in negative emotions such as Anger and Disgust. While few-shot prompting with Qwen-2.5-72B improves performance by 15% in the Disgust class compared to zero-shot prompting, it results in a 5% decrease in the Anger class. This trade-off suggests a potential semantic overlap between expressions of anger and disgust in the dataset, leading the model to occasionally misclassify nuanced emotional cues. This trade-off suggests a potential semantic overlap between expressions of anger and disgust in the dataset, leading the model to occasionally misclassify nuanced emotional cues. This trade-off suggests a potential semantic overlap between expressions of anger and disgust in the dataset, leading the model to occasionally misclassify nuanced emotional cues. But all other emotion classes (Joy, Fear, Sadness, Surprise) show significant improvement of more than 5% than their zero-shot equivalence. We also notice that the Llama-3.2-3B, Mistral-V3-7B, and



(a) Zero-shot setting

(b) Few-shot setting

Fig 11. Class-wise emotion recognition performance for two prompt settings across LLMs.



(a) Zero-shot setting

(b) Few-shot setting

Fig 12. Class-wise hate speech detection performance for two prompt settings across LLMs.

DeepSeek-R1-8B perform poorly across most of the categories, especially in Anger, Disgust, and Surprise, with both prompting approaches.

LLMs Performance Across Hate and Not-Hate Categories

Fig. 12 presents the class-wise F1-scores for the Hate Speech Detection task under both zero-shot and few-shot prompting settings. In the zero-shot setting, most models, including LLaMA-3.2-3B, struggle to identify the hate class accurately. The results show that several smaller or less multilingual models obtain an F1-score below 50% for the hate class, while their performance on the not-hate class remains disproportionately high. This pattern suggests that these models tend to prefer safer non-hateful predictions when no labelled examples are provided. Such imbalance indicates that smaller models may have limited ability to capture socially sensitive and context-dependent Bengali expressions. Similar challenges have been observed in other low-resource languages, where limited representation in pretraining data restricts a model's ability to grasp culturally grounded expressions.

Conversely, larger models handle the zero-shot setting more effectively. Under zero-shot prompting, Qwen-2.5-72B achieves an F1-score of 80.6% for the hate class and 79.2% for the not-hate class. Similarly, Gemma-2-27B records 79.2% for the hate class and 73.1% for the not-hate class.

In the few-shot setting, models that initially showed weak or moderate performance, such as LLaMA-3.2-3B and Phi-4-14B, demonstrate clear improvements in macro F1-score. These results indicate that few-shot prompting helps models detect both explicit and implicit hate expressions while maintaining more balanced classification. However, for larger multilingual models such as Qwen-2.5-72B and Gemma-2-27B, the performance gap between zero-shot and few-shot settings is relatively small, suggesting that these models already possess stronger contextual understanding before additional examples are provided.

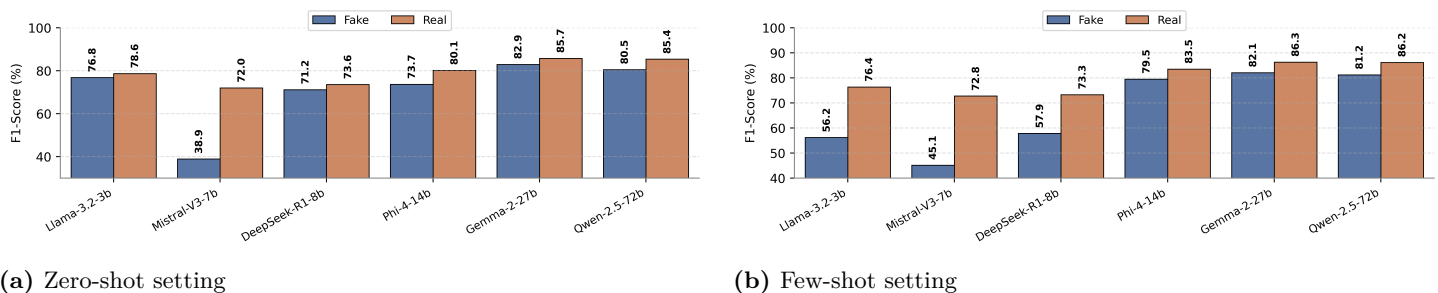


Fig 13. Class-wise fake news detection performance for two prompt settings across LLMs.

LLMs Performance Across Fake and Real News

Fig.13 presents the class-wise F1-scores for the Fake News Detection task under both zero-shot and few-shot prompting conditions. Under zero-shot setting, most models obtain the F1-score over 70% in both real and fake classes, which demonstrates they can generalise well without being access to any example with labels. Nonetheless, Mistral-V3-7B is particularly weak at detecting fake news, with an F1-score of only 38.9% for the fake class. The bigger guys, Gemma-2-27B and Qwen-2.5-72B to have the best performance in both classes, with more than 80% for each. This result demonstrates their superior performance in discriminating between real and fake content using the Bengali language.

Interestingly, the few-shot setting doesn't give an obvious improvement for this task. Smaller by sizes are models, such as DeepSeek-R1-8B obtain slight improvements (while LLaMA-3.2-3B actually performs worse), while the larger models have minor performance decrements. For instance, Gemma-2-27B exhibits F1-scores of 82.1% and 83.5%, in the case of the fake and real classes consecutively, versus 82.9% and 85.7%, obtained under the zero-shot setup. Similarly, Qwen-2.5-72B shows a marginal increase from 80.5% to 81.2% for the fake class and from 85.4% to 86.2% for the real class. This trend hints that, differently from other tasks, presenting few-shot instances does not largely contribute to the improvement of fake news detection. One possible explanation is that the models already have strong contextual and factual reasoning ability from pretraining, which renders further examples less helpful.

Error Analysis

This section analyzes the prediction behavior of the evaluated models using confusion matrices. The objective is to understand the common misclassification patterns and assess how model architecture and prompting strategies affect performance across the four Bengali text classification tasks.

Best Performing Models

The confusion matrices in Fig. 14 show the performance of the best-performing models. In the sentiment analysis task, Gemma-2-27B is competitive in both few-shot and zero-shot settings. As shown in Fig. 14(a) and Fig. 14(b), most samples align along the diagonal, indicating accurate predictions for positive and negative sentiments. A few neutral samples are misclassified as negative, which means that sometimes the model fails to detect mild form of neutrality in the colloquial Bengali text.

For the emotion recognition task, Qwen-2.5-72B demonstrates consistent reliability across major emotion categories, as seen in Fig. 14(c) and Fig. 14(d). The model is successful in recognising joy and fear, but the identification of anger and disgust

appears not to be as straightforward, possibly due to the similar linguistic cues they typically share. Virtually grounded prompting does give a small boost in performance for rare emotions such as surprise, which suggests that there is some benefit to seeing a small number of examples from the context.

In the hate speech detection task, Fig. 14(e) and Fig. 14(f) show that Qwen-2.5-72B achieves balanced classification between hate and not-hate categories. The few-shot setting improves recall for the hate class, helping the model detect implicit or context-dependent hostility expressed in Bengali. In the fake news detection task, shown in Fig. 14(g) and Fig. 14(h), Gemma-2-27B achieves high precision for both fake and real content with minimal off-diagonal errors. This consistency demonstrates that the model effectively uses linguistic and contextual cues to separate factual information from misinformation.

Overall, the best-performing models display clear diagonal concentration with very few misclassifications, confirming their strong contextual and linguistic understanding of Bengali text. Their advantage is likely due to larger parameter counts and extensive multilingual pre-training, which improve their adaptability to low-resource languages.

Worst Performing Models

Fig. 15 presents the confusion matrices for the least effective models. In sentiment analysis, shown in Fig. 15(a) and Fig. 15(b), LLaMA-3.2-3B and Mistral-V3-7B display significant confusion between neutral and negative samples, with weak diagonal concentration. This suggests that smaller models lack sufficient semantic depth to capture subtle tonal shifts in Bengali sentiment expressions.

For emotion recognition, Fig. 15(c) and 15(d) show that Mistral-V3-7B and DeepSeek-R1-8B produce scattered distributions with frequent overlaps between sadness, disgust, and anger. The few-shot configuration yields limited improvement, suggesting that model capacity, rather than exposure to examples, limits emotional differentiation..

In the hate speech detection task (Fig. 15(e) and 15(f)), both DeepSeek-R1-8B and LLaMA-3.2-3B misclassify many hateful statements as not-hate. [This pattern suggests that the models may be relatively conservative when identifying hate speech, leading to a higher number of false negatives. However, the underlying causes of these errors cannot be conclusively determined from the confusion matrices alone and would require further qualitative error analysis.](#) Finally, for fake news detection, shown in Fig. 15(g) and Fig. 15(h), Mistral-V3-7B struggles to detect fabricated content, frequently identifying fake instances as real. These scattered prediction patterns demonstrate the limited generalization capability of smaller or less optimised models.

Qualitative Linguistic Error Analysis

[While confusion matrices provide a quantitative overview of model performance, a manual examination of misclassified instances revealed several recurring linguistic challenges in Bengali text classification.](#)

[First, the models frequently struggled with sarcasm and implicit sentiment. For example, expressions containing seemingly positive lexical items were often used sarcastically, causing the models to rely on surface-level word associations rather than the intended pragmatic meaning. This phenomenon was particularly evident in the Emotion Recognition and Sentiment Analysis tasks, where sarcastic or ironic statements were commonly confused with positive emotions.](#)

[Second, errors were observed in sentences containing negation within otherwise positive contexts. Bengali speakers often use constructions such as “তর মতো মানুষ হয় না” to express strong admiration, despite the presence of a negative marker. Several](#)

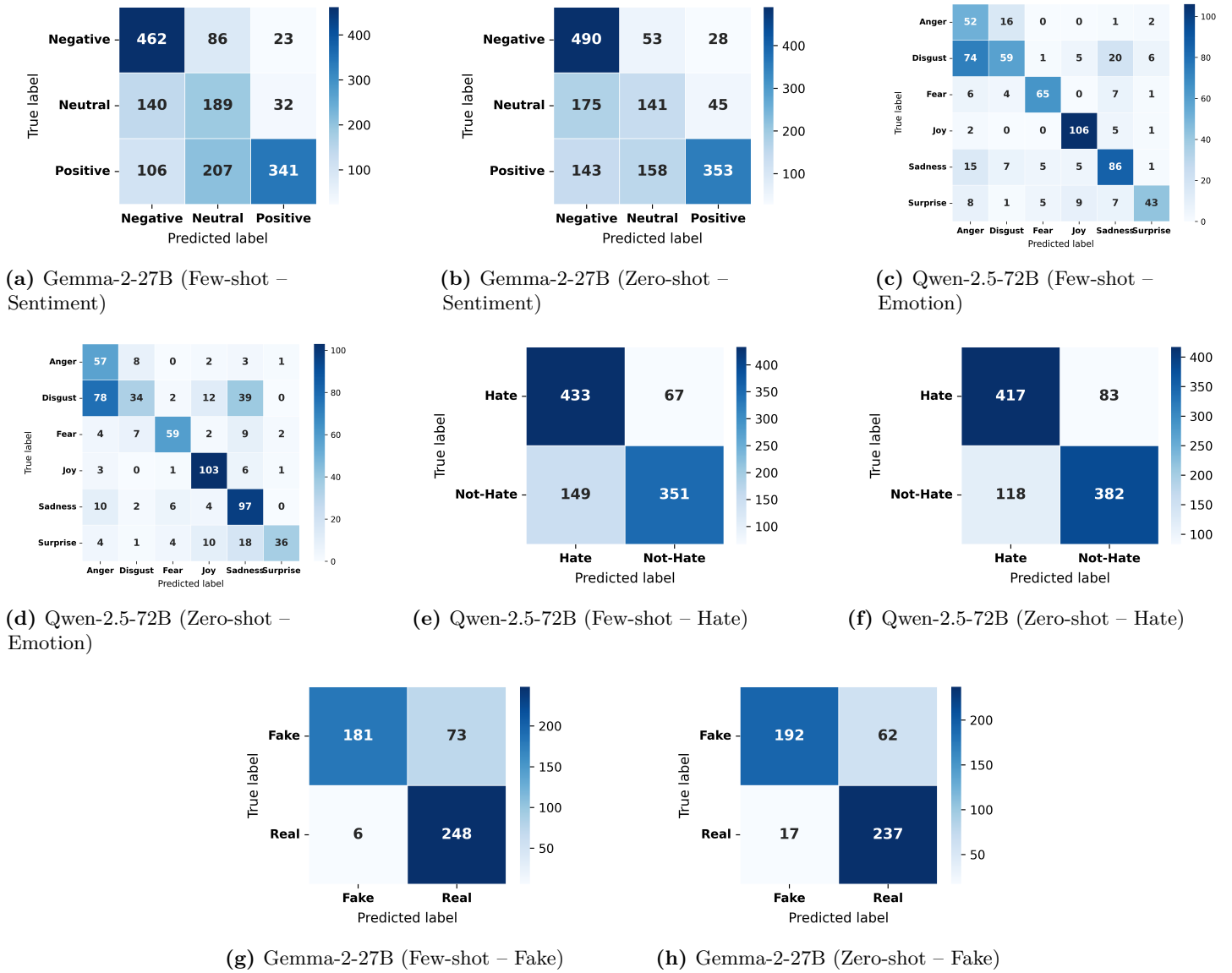


Fig 14. Confusion matrices of top LLMs on Bengali text classification tasks under zero- and few-shot settings.

models incorrectly interpreted such statements as negative sentiment, suggesting difficulties in capturing the interaction between negation and contextual meaning.

Third, the Hate Speech Detection task revealed challenges related to informal slang and offensive language. Certain colloquial expressions and mild insults were frequently classified as hate speech despite being used in non-hateful conversational contexts. Conversely, some implicitly abusive or derogatory statements were misclassified as non-hateful because they lacked explicit offensive keywords. These observations suggest that the models often relied on lexical cues rather than broader contextual interpretation.

Fourth, in the Fake News Detection task, many fabricated articles were written using realistic journalistic structures, including named entities, numerical details, quotations, and formal news-reporting language. As a result, several fake news instances were incorrectly classified as real news because the models appeared to rely

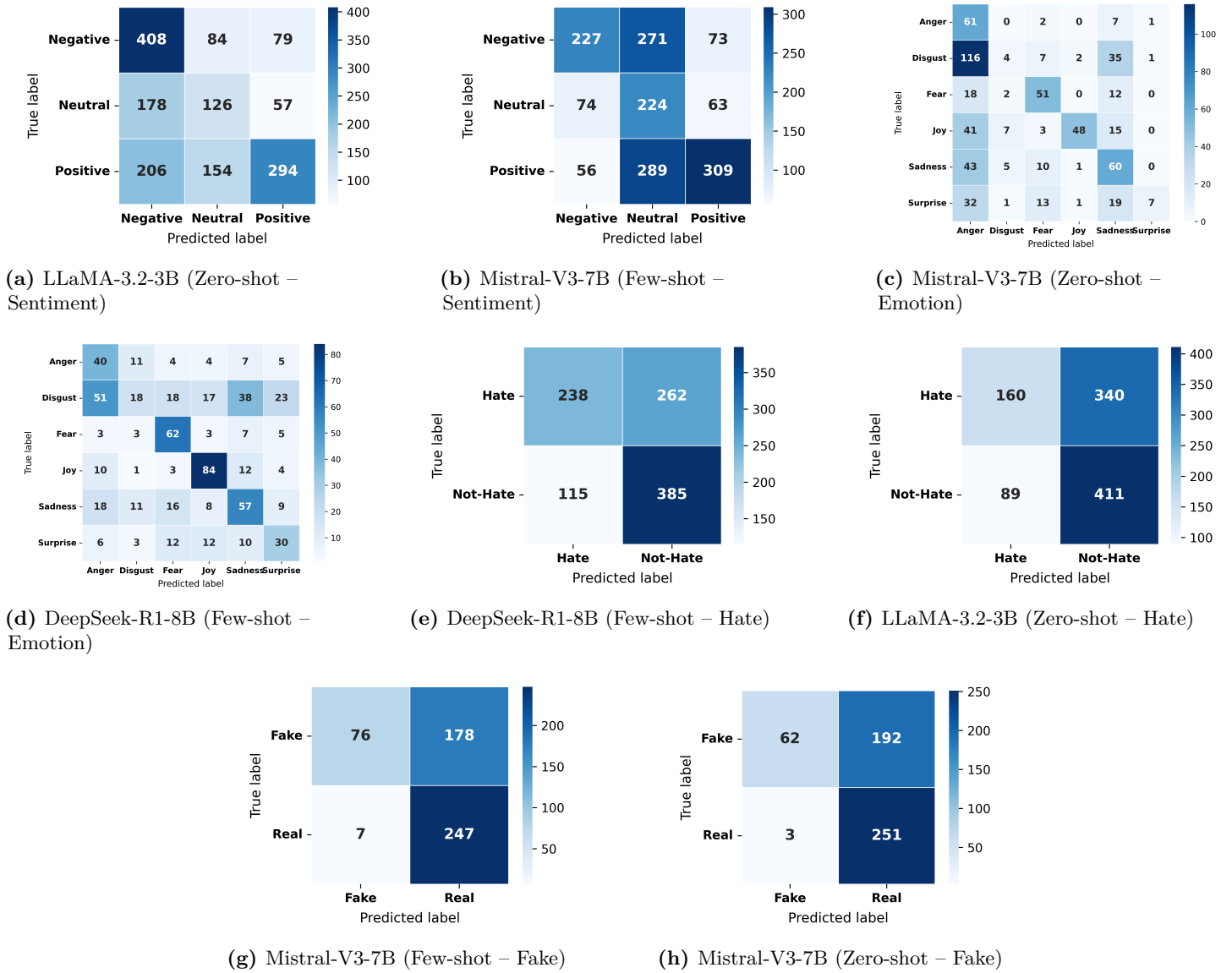


Fig 15. Confusion matrices of low-performing LLMs showing misclassification patterns in Bengali tasks.

on surface-level credibility cues rather than verifying factual plausibility. This finding suggests that stylistically convincing misinformation remains challenging even for large-scale LLMs.

Overall, the qualitative analysis indicates that pragmatic language phenomena, including sarcasm, contextual negation, culturally specific slang usage, and the realistic journalistic style of fabricated news articles, constitute important sources of error for current open-source LLMs in Bengali text classification. Future work may conduct a more fine-grained linguistic investigation to systematically quantify the impact of these phenomena across different tasks and model architectures.

Receiver Operating Characteristic (ROC-AUC) Analysis

In this section, we evaluate the discriminative performance of the models using confusion matrix analysis, ROC curves, and AUC scores. The ROC-AUC score

attempts to present a fair judgment to compare models’ ability to distinguish between classes across different prompting strategies. For multiclass classification tasks (e.g., Sentiment Analysis, Emotion Recognition), the One-vs-Rest (OvR) strategy is employed to calculate the area under the ROC curve (AUC), where each class is iteratively treated as the positive class against all others. Fig. 16 illustrates the ROC–AUC results for all four Bengali text classification tasks under both zero-shot and few-shot conditions.

As shown in Fig. 16(a) and Fig. 16(b), the sentiment analysis task reveals poor to acceptable discrimination for the evaluated models, with AUC values ranging from 0.63 to 0.72. Gemma-2-27B achieves the highest AUC of 0.72 in the few-shot setting, followed closely by Qwen-2.5-72B (0.70), demonstrating more confident and stable classification boundaries. Smaller models such as LLaMA-3.2-3B and Mistral-V3-7B show weaker discrimination, indicating difficulties in capturing subtle sentiment cues in Bengali.

For the emotion recognition task (Fig. 16(c) and Fig. 16(d)), all models get a higher AUC value than that of the sentiment task. As shown in the few-shot setting, Qwen-2.5-72B and Gemma-2-27B achieve higher 0.81 and 0.80 than others, revealing their healthy discriminative power on emotional tone. While Phi-4-14B also maintains a competitive AUC at 0.77, smaller models fail to consistently capture sensitivity between groups of emotion classes.

In hate speech detection (Fig. 16(e) and Fig. 16(f)), Gemma-2-27B and Qwen-2.5-72B outperform the other ones with few-shot AUCs higher than 0.74 and 0.78, respectively. This improvement highlights the benefit of in-context learning for sensitive social classification tasks where context and tone are important factors. By contrast, LLaMA-3.2-3B gets a low AUC of only 0.57 in the zero-shot setting, indicating that LLaMA-3.2-3B is less strong in detecting offensive or hateful content.

The fake news detection task (Fig. 16(g) and Fig. 16(h)) obtains the highest AUC values in all tasks. Both Gemma-2-27B and Qwen-2.5-72B obtain 0.84 in the two versions, prompting, indicating a good discriminative level for real and fake text. This consistent line of results further confirms that the larger multilingual LLMs are indeed more capable to identify misinformation due to their access to a wider range of facts and contexts.

Discussion

This section provides a summary based on the observations and errors of the results, discussing some of the trends related to performance in all tasks across various LLMs under zero-shot and few-shot prompting. The tendency of the general performance is shown in Fig. 17. The easiest task among those analysed is Fake News detection, which reaches high F1-scores in all models regardless of the dataset used; Emotion Recognition remains as the toughest one. Overall, few-shot prompting significantly enhances model performance, especially in Emotion and Hate Speech detection, emphasising the significance of contextualised learning. In addition, larger models like Qwen-2.5-72B and Gemma-2-27B are uniformly superior on all tasks. While Qwen-2.5-72B leverages its extensive multilingual pretraining, Gemma-2-27B demonstrates that sheer model scale and architectural efficiency can compensate for a pretraining corpus primarily focused on English.

Small models Mistral-V3-7B, Phi-4-14B, and DeepSeek-R1-8B perform relatively less well in general on the all tasks, especially under zero-shot. Their relatively low performance might result from a smaller number of parameters and the amount of languages they were less exposed to when being pretrained. Models pre-trained on English (or other high-resource languages) often overlook unimodal and linguistic

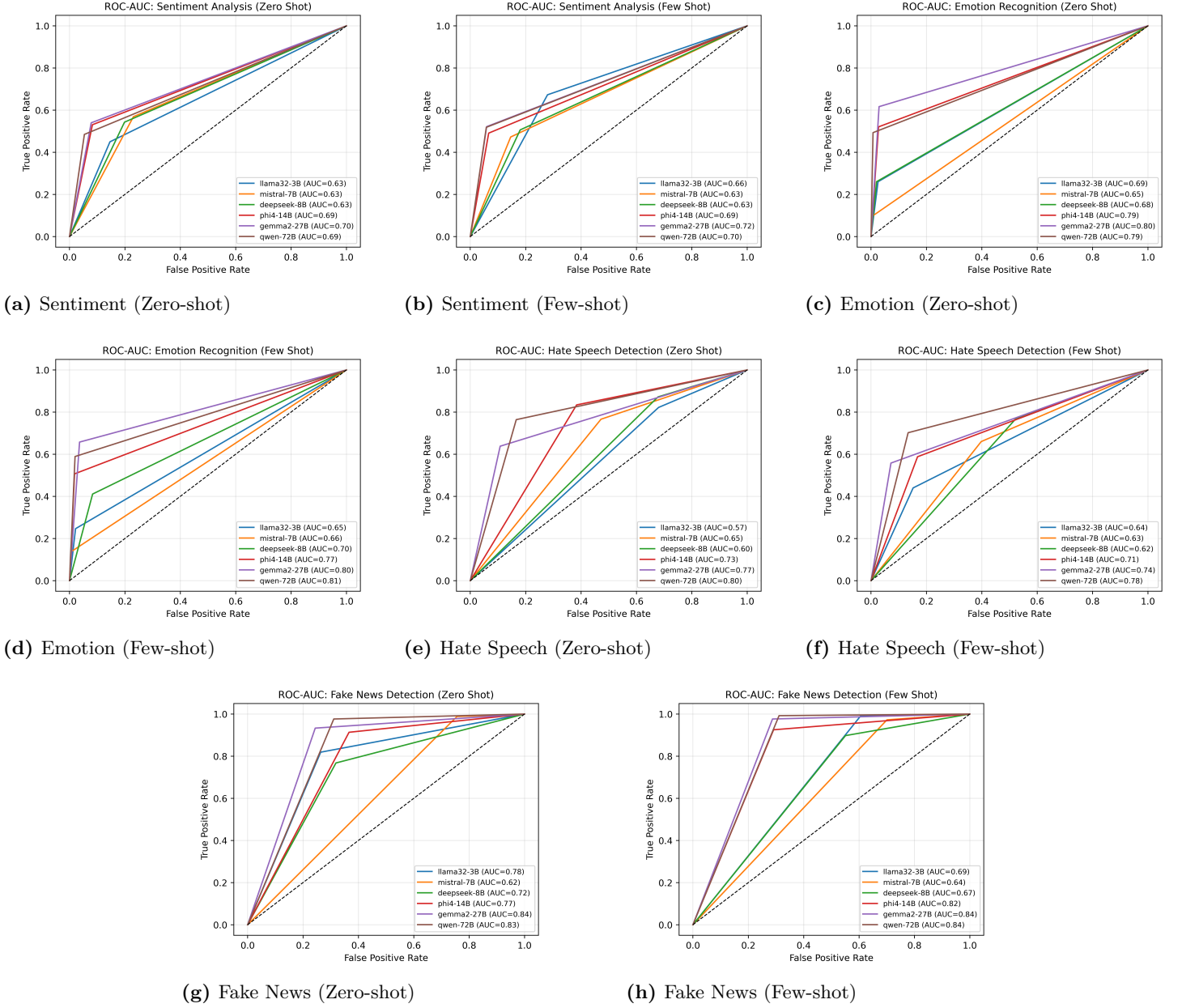
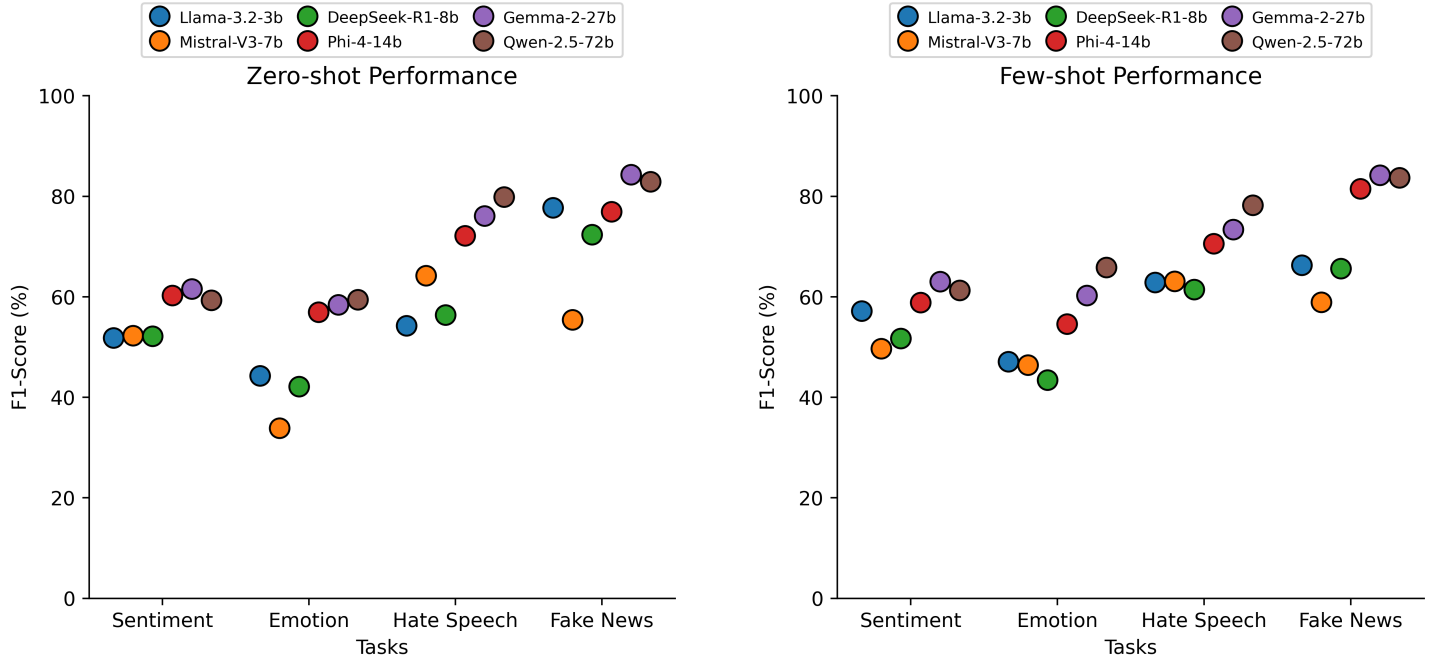


Fig 16. ROC-AUC curves of six LLMs on Bengali text classification tasks under zero- and few-shot settings.

conditions, missing out because of the absence of coverage for low-resource languages, as well as less-than-desirable syntactic or cultural alignment. This highlights the importance of scale and multilingual data diversity for generalisation across low-resource languages. Successful performance of Qwen-2.5-72B and Gemma-2-27B is thus a testament to not only their higher capacity models but also the amount of pretraining they have received multilingually, enabling greater adaptability to morphologically rich languages like Bengali.

Differences in model behaviour are also evident at the task level. Sentiment analysis is relatively simple yet only achieves moderate results, i.e. best models obtain 62–63% F1 scores. The conundrum is how to distinguish neutral orientation from the



(a) Zero-shot setting

(b) Few-shot setting

Fig 17. LLMs' performance with two prompt settings across various classification tasks.

negative one, as could be seen in the confusion matrix result. It should also be noted that the SentiNoB dataset exhibits class imbalance, containing 654 positive samples, 571 negative samples, and only 361 neutral samples. Therefore, the relatively lower F1-scores observed for the neutral class may be influenced not only by model limitations but also by the unequal class distribution. The reduced representation of neutral samples may make it more difficult for LLMs to accurately identify the characteristics of this category. Consequently, class imbalance should be considered a potential contributing factor when interpreting neutral-class performance. The emotion recognition still remains the most challenging task, while the state-of-the-art method yields 65.76% F1-score in the few-shot scenario. This limitation might be due to the fine-grained nature of our task where we have to discriminate between six very similar feelings that often share overlapping characteristics in Bengali.

Results of hate speech detection are more promising, with state-of-the-art approaches approaching 80% F1-score. The binary nature of the task, and better linguistic cues for hate speech make it relatively easier. However, the error analysis also indicates a small over-prediction as compared to the hateful labels that may imply a conservative classification approach, good for moderation, but prone to raising false positives. Fake News detection is the most tractable task, with the best-performing models achieving F1-scores over 84%. This high accuracy is most likely due to the clear stylistic and factual differences between real and fake articles.

Few-shot learning effect is task-dependent. It proves effective for sentiment analysis, hate speech detection, and emotion recognition consistently, while its impact is constrained or slightly negative regarding fake news detection. This tendency indicates that fake news detection benefits mostly from previous knowledge of facts and pretrained representations, rather than from abundant in-context examples. On the other hand, ones requiring fine-grained tones or affects such as sentiment or emotion classification benefit significantly from just a few labelled examples; this

provides strong evidence for the importance of in-context learning in linguistic subtlety under low-resource settings.

A further consideration concerns the absence of comparisons with Bengali-specific pre-trained language models, such as BanglaBERT, and proprietary frontier LLMs, such as GPT-4. Recent studies have emphasized the importance of incorporating strong baseline models to provide a more comprehensive and rigorous evaluation of proposed approaches [51–55]. In the present study, our primary objective was to conduct a systematic comparison of contemporary open-source instruction-tuned LLMs under identical zero-shot and few-shot prompting settings. Consequently, the experimental design focused exclusively on publicly available open-source models. Furthermore, BanglaBERT is typically evaluated under supervised fine-tuning settings, whereas our work investigates the in-context learning capabilities of generative LLMs without task-specific parameter updates. Therefore, a direct comparison would not constitute a fully equivalent evaluation setting. Nevertheless, we acknowledge that including Bengali-specific encoder models and proprietary frontier LLMs would provide additional insights into the relative strengths and limitations of open-source LLMs. Future work should incorporate such baselines to further strengthen benchmarking efforts for Bengali NLP tasks.

In addition to these discoveries, the analysis yields other overarching conclusions. Recognizing emotions is still the hardest thing to do because many Bengali ways of showing emotions are quite peculiar to their culture or situation. In real conversation, the lines between feelings like anger, disgust, and melancholy are typically very thin, which makes it harder for models to tell them apart without clear instances. On the other hand, Qwen-2.5-72B and Gemma-2-27B consistently perform well. Qwen-2.5-72B benefits from superior multilingual pretraining and better representation of Indic scripts. While Gemma-2-27B was trained primarily on English data and not explicitly optimized for multilingual state-of-the-art performance, its sheer scale and architectural efficiency still allow it to perform robustly on Bengali tasks. These traits let them pick up on morphological changes, compound forms, and subtle semantic differences that smaller models don’t always catch. More broadly, the results show that there are some important rules for low-resource NLP: model scale alone isn’t enough; it’s important to have a variety of multilingual data; in-context examples are especially useful for tasks that involve fine-grained linguistic signals; and balanced evaluation datasets are important for fair comparison across LLMs. These results show that Bengali NLP will only get better in the future if there are more language-specific resources and better multilingual training pipelines.

Implications for Low-Resource Language NLP

Findings from this research study carry several crucial results for furthering NLP in languages with low-resources like Bengali. First, the wide gap in performance between large-scale models (Qwen-2.5-72B and Gemma-2-27B) and small ones (LLaMA-3.2-3B and Mistral-V3-7B) proves that large model sizes continue to play a decisive role in dealing with the arithmetic diversity and morphological complexity of languages. But this also results in an accessibility challenge, as it remains a significant obstacle for many researchers and institutions working in resource-constrained environments the computational and hardware requirements of big models.

Second, the consistent effectiveness of few-shot prompting on most tasks shows that meaningful performance gains can still be achieved with limited labeled data. This finding is particularly valuable for languages without large annotated datasets, where creating large corpora is difficult or expensive. With the creation of small but high-quality labelled datasets, future research in Bengali and other similar languages should move away from only looking at developing huge corpora. These datasets are

well-suited to few-shot learning, allowing LLMs to infer effectively with few examples whilst keeping computational demands limited.

Thirdly, differences in task difficulty demonstrate that the choice of method and problem framing significantly affect model performance. Tasks with clear-cut categorical distinctions—e.g., hate speech and fake news detection can be quite directly solved using our existing LLM architectures. In contrast, fine-grained tasks like emotion recognition, which require thinking deeply about subtle expressions and environmental cues, might profit from different methods such as fine-tuning, ensemble learning or hybrids of LLMs with task-specific classifiers. The development of adaptive methods that balance generalisation against linguistic sensitivity could make models more robust in complex low-resource language environments.

Limitations

While this study provides valuable insights into the performance of open-source LLMs for Bengali text classification, several limitations must be acknowledged. First, our evaluation is restricted exclusively to classification tasks. Although classification is a fundamental component of NLP, future research should expand this analysis to include generative tasks such as text summarization, machine translation, and question answering to draw more comprehensive conclusions about LLM capabilities in Bengali.

In addition, for easier transparency and reproducibility, the research will only use publicly available models. However, this selection excludes the comparison to closed-source proprietary systems (such as GPT-4 or Claude), which might have a better overall performance. They should make it to future benchmarks — providing a better perspective of how wide the open-closed ecosystem gap is for Bengali language processors.

Third, the prompting method adopted here is exclusive to zero-shot and few-shot (five-shot) settings. More advanced methods of prompting (e.g., chain-of-thought reasoning, self-consistency or retrieval-augmented generation) could lead to more improvements but were not in the scope of this work. Also, parameter-efficient fine-tuning mechanisms such as LoRA or QLoRA, could be pursued and provide practical ways to improve the performance of smaller models without spending much in terms of computing time.

Fourth, the datasets we used, while standard benchmarks in the Bengali NLP community, suffer from natural limitations in terms of size and coverage, domain diversity, and annotation consistency. For instance, much of the data originates from social media platforms and may not accurately reflect formal or domain-specific Bengali. Second, some datasets are small (e.g., the emotion recognition corpus: only 625 examples), which may compromise statistical stability and the generalisation value of the results. Furthermore, the SentiNoB dataset exhibits class imbalance, containing 654 positive samples, 571 negative samples, and only 361 neutral samples. This imbalance may have influenced class-wise performance, particularly the lower F1-scores observed for the neutral category in the sentiment analysis task.

Finally, although we focus on quantitative performance measures (i.e., accuracy or F1-score) in this work, we do not delve into more holistic notions of fairness, bias, and cultural relevance of model predictions. Because of the sensitive nature of tasks like hate speech detection and fake news classification, future work must also include bias assessment frameworks to analyse fine-grained model behaviour in different demographic groups, social context, and dialectic variation within the Bengali-speaking community.

Conclusion

A comprehensive evaluation of six open-source Large Language Models (LLMs)—LLaMA-3.2-3B, Mistral-V3-7B, DeepSeek-R1-8B, Phi-4-14B, Gemma-2-27B, and Qwen-2.5-72B is presented with this research, where four Bengali text classification tasks are evaluated, such as sentiment analysis, emotion recognition, hate-speech detection, and fake-news detection. For ensuring equal and consistent comparisons, both zero-shot and few-shot prompting strategies were applied along with vLLM. The outcome of this research is that it sheds light on the fact that the larger multilingual models constantly surpass smaller ones across all tasks. With a zero-shot setting, the best-performing model on hate-speech detection is Qwen-2.5-72B with 79.90% accuracy and a **Macro-F1** score of 79.88%, whereas for fake-news classification, 84.45% accuracy is achieved with Gemma-2-27B as well as an 84.33% **Macro-F1** score. After a few-shot prompt was applied, Gemma-2-27B maintained strong performance on fake-news detection with a **Macro-F1** score of 84.17%. Qwen-2.5-72B achieved the best emotion-recognition result, reaching a **Macro-F1** score of 78.25%. These results suggest that the size of the model, the quality of multilingual pretraining, and the use of in-context examples all play an important role in improving performance for low-resource languages such as Bengali.

Future Work

Though the findings are pretty favourable, several issues still need careful attention. A difficulty is seen among the models that are working to classify the neutral as well as the negative statement, and they also frequently lump together emotions with similar meanings, such as anger and disgust, as well as misleading statements, which are also labelled as real. These mechanisms can be strengthened further in the future. For improving the ability of the model, approaches that are improved should be applied, such as step-by-step reasoning or the use of external information sources. With an intention to improve the performance, along with the requirement of not using expensive hardware, Lightweight fine-tuning approaches such as LoRA or QLoRA can be applied. In order to acquire an improved awareness of their overall capabilities and limits, a broader set of NLP tasks, for instance, text generation, summarisation, and question answering, should be tested as well. Additionally, retaining the NLP system, which is a blessing for Bangali users, should pay great attention to its fairness across different social and demographic groups. With a view to improving further, larger and more diverse datasets should be introduced, which could be a great help to Bangali NLP tool users.

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Data Availability Statement: The datasets and code used in this study are publicly available at: 1148

<https://github.com/hamidhosen42/Bengali-MultiDomain-LLM-Evaluation>. 1149

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